III. Introduction to Drug Classes

The Controlled Substances Act (CSA) regulates five classes of drugs:

- Narcotics
- Depressants
- Stimulants
- Hallucinogens
- Anabolic steroids

Each class has distinguishing properties, and drugs within each class often produce similar effects. However, all controlled substances, regardless of class, share a number of common features. This introduction will familiarize you with these shared features and define the terms frequently associated with these drugs.

All controlled substances have abuse potential or are immediate precursors to substances with abuse potential. With the exception of anabolic steroids, controlled substances are abused to alter mood, thought, and feeling through their actions on the central nervous system (brain and spinal cord). Some of these drugs alleviate pain, anxiety, or depression. Some induce sleep and others energize.

Though some controlled substances are therapeutically useful, the "feel good" effects of these drugs contribute to their abuse. The extent to which a substance is reliably capable of producing intensely pleasurable feelings (euphoria) increases the likelihood of that substance being abused.

DRUG ABUSE

When controlled substances are used in a manner or amount inconsistent with the legitimate medical use, it is called drug abuse. The non-sanctioned use of substances controlled in Schedules I through V of the CSA is considered drug abuse. While legal pharmaceuticals placed under control in the CSA are prescribed and used by patients for medical treatment, the use of these same pharmaceuticals outside the scope of sound medical practice is drug abuse.

DEPENDENCE

In addition to having abuse potential, most controlled substances are capable of producing dependence, either physical or psychological.

Physical Dependence

Physical dependence refers to the changes that have occurred in the body after repeated use of a drug that necessitates the continued administration of the drug to prevent a withdrawal syndrome. This withdrawal syndrome can range from mildly unpleasant to life-threatening and is dependent on a number of factors, such as:

- The drug being used
- The dose and route of administration
- Concurrent use of other drugs
- Frequency and duration of drug use
- The age, sex, health, and genetic makeup of the user

Psychological Dependence

Psychological dependence refers to the perceived "need" or "craving" for a drug. Individuals who are psychologically dependent on a particular substance often feel that they cannot function without continued use of that substance. While physical dependence disappears within days or weeks after drug use stops, psychological dependence can last much longer and is one of the primary reasons for relapse (initiation of drug use after a period of abstinence).

Contrary to common belief, physical dependence is not addiction. While individuals with a substance use disorder are usually physically dependent on the drug they are abusing, physical dependence can exist without addiction. For example,

patients who take narcotics for chronic pain management or benzodiazepines to treat anxiety are likely to be physically dependent on that medication.

ADDICTION

Addiction is defined as compulsive drug-seeking behavior where acquiring and using a drug becomes the most important activity in the user's life. This definition implies a loss of control regarding drug use, and the person with a substance use disorder will continue to use a drug despite serious medical and/or social consequences. In 2015, an estimated 27.1 million Americans aged 12 or older were current (past month) illicit drug users, meaning they had used an illicit drug during the month prior to the survey interview. This estimate represents 10.1 percent of the population aged 12 or older. Illicit drugs include marijuana, cocaine (including crack), heroin, hallucinogens, inhalants, methamphetamine, or prescription psychotherapeutics (including pain relievers, tranquilizers, stimulants, and sedatives) that were misused. [1]

Drugs within a class are often compared with each other with terms like potency and efficacy. Potency refers to the amount of a drug that must be taken to produce a certain effect, while efficacy refers to whether or not a drug is capable of producing a given effect regardless of dose. Both the strength and the ability of a substance to produce certain effects play a role in whether that drug is selected by the drug user.

It is important to keep in mind that the effects produced by any drug can vary significantly and is largely dependent on the dose and route of administration. Concurrent use of other drugs can enhance or block an effect, and substance abusers often take more than one drug to boost the desired effects or counter unwanted side effects. The risks associated with drug abuse cannot be accurately predicted because each user has his/her own unique sensitivity to a drug. There are a number of theories that attempt to explain these differences, and it is clear that a genetic component may predispose an individual to certain toxicities or even addictive behavior.

Youth are especially vulnerable to drug abuse. According to the National Institute on Drug Abuse, young Americans engaged in extraordinary levels of illicit drug use in the last third of the twentieth century. Today, about 48 percent of young people have used an illicit drug by the time they leave high school and about 7 percent of eighth graders, 16 percent of tenth graders, and 24 percent of twelfth graders are current (within the past month) users. [2]

Substance abuse in youth can result in tragic consequences with untold harm to themselves, their families, and others. The 2016 Surgeon General's Report on Alcohol, Drugs, and Health identified risk factors for youth which might lead them into substance abuse. These include being raised in a home where the parents or other relatives use drugs, living in neighborhoods and going to schools where drug use is common, and associating with peers who use substances. Nearly 70 percent of those who try an illicit drug before the age of 13 develop a substance use disorder in the next 7 years, compared with 27 percent of those who first try an illicit drug after the age of 17. [3]

In the sections that follow, each of the five classes of drugs is reviewed and various drugs within each class are profiled. Although marijuana is classified in the CSA as a hallucinogen, a separate section is dedicated to that topic. There are also a number of substances that are abused but not regulated under the CSA. Alcohol and tobacco, for example, are specifically exempt from control by the CSA. In addition, a whole group of substances called inhalants are commonly available and widely abused by children. Control of these substances under the CSA would not only impede legitimate commerce, but also would likely have little effect on the abuse of these substances by youngsters. An energetic campaign aimed at educating both adults and youth about inhalants is more likely to prevent their abuse. To that end, a section is dedicated to providing information on inhalants.

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[1] *Results from the 2015 National Survey on Drug Use and Health: Detailed Tables; U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration*

2 *Monitoring the Future Survey, 2016; National Institute on Drug Abuse, National Institutes of Health, U.S. Department of Health and Human Services*

3 *Facing Addiction in America. The Surgeon General's Report on Alcohol, Drugs, and Health, October 2016. U.S. Department of Health and Human Services.*

IV. Narcotics

WHAT ARE NARCOTICS?

Also known as "opioids," the term "narcotic" comes from the Greek word for "stupor" and originally referred to a variety of substances that dulled the senses and relieved pain. Though some people still refer to all drugs as "narcotics," today "narcotic" refers to opium, opium derivatives, and their semi-synthetic substitutes. A more current term for these drugs, with less uncertainty regarding its meaning, is "opioid." Examples include the illicit drug heroin and pharmaceutical drugs like OxyContin, Vicodin, codeine, morphine, methadone, and fentanyl.

WHAT IS THEIR ORIGIN?

The poppy Papaver somniferum is the source for all natural opioids, whereas synthetic opioids are made entirely in a lab and include meperidine, fentanyl, and methadone. Semi-synthetic opioids are synthesized from naturally occurring opium products, such as morphine and codeine, and include heroin, oxycodone, hydrocodone, and hydromorphone. Teens can obtain narcotics from friends, family members, medicine cabinets, pharmacies, nursing homes, hospitals, hospices, doctors, and the Internet.



OxyContin 160 mg tablet



Heroin

What are common street names?

Street names for various narcotics/opioids include:

- Smack, Horse, Mud, Brown Sugar, Junk, Black Tat, Big H, Paregoric, Dover's Powder, MPTP (New Heroin), Hilbilly Heroin, Lean or Purple Drank, OC, Ox, Oxy, Oxycotton, Sippin Syrup

What do they look like?

Narcotics/opioids come in various forms, including:

- Tablets, capsules, skin patches, powder, chunks in varying colors (from white to shades of brown and black), liquid form for oral use and injection, syrups, suppositories, and lollipops

How are they abused?

- Narcotics/opioids can be swallowed, smoked, sniffed, or injected.

What is their effect on the mind?

Besides their medical use, narcotics/opioids produce a general sense of well-being by reducing tension, anxiety, and aggression. These effects are helpful in a therapeutic setting but contribute to the drugs' abuse. Narcotic/opioid use comes with a variety of unwanted effects, including drowsiness, inability to concentrate, and apathy.

Psychological dependence

Use can create psychological dependence. Long after the physical need for the drug has passed, the addict may continue to think and talk about using drugs and feel overwhelmed coping with daily activities. Relapse is common if there are not changes to the physical environment or the behavioral motivators that prompted the abuse in the first place.

What is their effect on the body?

Narcotics/opioids are prescribed by doctors to treat pain, suppress cough, cure diarrhea, and put people to sleep. Effects depend heavily on the dose, how it's taken, and previous exposure to the drug. Negative effects include:

- Slowed physical activity, constriction of the pupils, flushing of the face and neck, constipation, nausea, vomiting, and slowed breathing

As the dose is increased, both the pain relief and the harmful effects become more pronounced. Some of these preparations are so potent that a single dose can be lethal to an inexperienced user. However, except in cases of extreme intoxication, there is no loss of motor coordination or slurred speech.

Physical dependence and withdrawal

Physical dependence is a consequence of chronic opioid use, and withdrawal takes place when drug use is discontinued. The intensity and character of the physical symptoms experienced during withdrawal are directly related to the particular drug used, the total daily dose, the interval between doses, the duration of use, and the health and personality of the user. These symptoms usually appear shortly before the time of the next scheduled dose.

Early withdrawal symptoms often include:

- Watery eyes, runny nose, yawning, and sweating

As the withdrawal worsens, symptoms can include:

- Restlessness, irritability, loss of appetite, nausea, tremors, drug craving, severe depression, vomiting, increased heart rate and blood pressure, and chills alternating with flushing and excessive sweating

However, without intervention, the withdrawal usually runs its course, and most physical symptoms disappear within days or weeks, depending on the particular drug.

What are their overdose effects?

Overdoses of narcotics are not uncommon and can be fatal. Physical signs of narcotics/opioid overdose include:

- Constricted (pinpoint) pupils, cold clammy skin, confusion, convulsions, extreme drowsiness, and slowed breathing

Which drugs cause similar effects?

With the exception of pain relief and cough suppression, most central nervous system depressants (like barbiturates, benzodiazepines, and alcohol) have similar effects, including slowed breathing, tolerance, and dependence.

What is their legal status in the United States?

Narcotics/opioids are controlled substances that vary from Schedule I to Schedule V, depending on their medical usefulness, abuse potential, safety, and drug dependence profile. Schedule I narcotics, like heroin, have no medical use in the U.S. and are illegal to distribute, purchase, or use outside of medical research.

Fentanyl



Fentanyl

WHAT IS FENTANYL?

Fentanyl is a potent synthetic opioid drug approved by the Food and Drug Administration for use as an analgesic (pain relief) and anesthetic. It is approximately 100 times more potent than morphine and 50 times more potent than heroin as an analgesic.

WHAT IS ITS ORIGIN?

Fentanyl was first developed in 1959 and introduced in the 1960s as an intravenous anesthetic. It is legally manufactured and distributed in the United States. Licit fentanyl pharmaceutical products are diverted via theft, fraudulent prescriptions, and illicit distribution by patients, physicians, and pharmacists.

From 2005 through 2007, both fatal overdoses associated with abuse of clandestinely produced fentanyl and law enforcement encounters increased markedly. According to the Centers for Disease Control and Prevention, there were 1,013 fatal overdoses recorded from April 2005 to March 2007. More recently, there has been a re-emergence of trafficking, distribution, and abuse of illicitly produced fentanyl with an associated dramatic increase in overdose fatalities.

What are common street names?

Common street names include:

Apache, China Girl, China Town, Dance Fever, Friend, Goodfellas, Great Bear, He-Man, Jackpot, King Ivory, Murder 8, and Tango & Cash.

What does it look like?

Fentanyl pharmaceutical products are currently available in the following dosage forms: oral transmucosal lozenges commonly referred to as fentanyl "lollipops" (Actiq), effervescent buccal tablets (Fentora), sublingual tablets (Abstral), sublingual sprays (Subsys), nasal sprays (Lazanda), transdermal patches (Duragesic), and injectable formulations.

Clandestinely produced fentanyl is encountered either as a powder or in counterfeit tablets and is sold alone or in combination with other drugs such as heroin or cocaine.

How is it abused?

Fentanyl can be injected, snorted/sniffed, smoked, taken orally by pill or tablet, and spiked onto blotter paper. Fentanyl patches are abused by removing its gel contents and then injecting or ingest-

ing these contents. Patches have also been frozen, cut into pieces, and placed under the tongue or in the cheek cavity. Illicitly produced fentanyl is sold alone or in combination with heroin and other substances and has been identified in counterfeit pills, mimicking pharmaceutical drugs such as oxycodone. According to the National Forensic Laboratory Information System, reports on fentanyl (both pharmaceutical and clandestinely produced) increased from nearly 5,400 in 2014 to over 14,600 in 2015, as reported by federal, state, and local forensic laboratories in the United States.

What is the effect on the body?

Fentanyl, similar to other commonly used opioid analgesics(e.g., morphine), produces effects such as relaxation, euphoria, pain relief, sedation, confusion, drowsiness, dizziness, nausea, vomiting, urinary retention, pupillary constriction, and respiratory depression.

What are the overdose effects?

Overdose may result in stupor, changes in pupillary size, cold and clammy skin, cyanosis, coma, and respiratory failure leading to death. The presence of triad of symptoms such as coma, pinpoint pupils, and respiratory depression are strongly suggestive of opioid poisoning.

Which drugs cause similar effects?

Drugs that cause similar effects include other opioids such as morphine, hydrocodone, oxycodone, hydromorphone, methadone, and heroin.

What is the legal status in the Federal Control Substances Act?

Fentanyl is a Schedule II narcotic under the United States Controlled Substances Act of 1970.

Heroin

WHAT IS HEROIN?

Heroin is a highly addictive drug and it is a rapidly acting opioid.

WHAT IS ITS ORIGIN?

Heroin is processed from morphine, a naturally occurring substance extracted from the seed pod of certain varieties of poppy plants grown in:

- Mexico, South America, Southwest Asia (Afghanistan and Pakistan), and Southeast Asia (Thailand, Laos, and Myanmar (Burma))

Heroin comes in several forms, primarily white powder from Mexico and South America; and "black tar" and brown powder from Mexico.

What are common street names?

Common street names for heroin include:

- Big H, Black Tar, Chiva, Hell Dust, Horse, Negra, Smack, and Thunder

What does it look like?

Heroin is typically sold as a white or brownish powder, or as the black sticky substance known on the streets as "black tar heroin." Although purer heroin is becoming more common, most street heroin is "cut" with other drugs or with substances such as sugar, starch, powdered milk, or quinine.

How is it abused?

Heroin can be injected, smoked, or sniffed/snorted. High purity heroin is usually snorted or smoked.

What is its effect on the mind?

Because it enters the brain so rapidly, heroin is particularly addictive, both psychologically and physically. Heroin users report feeling a surge of euphoria or "rush," followed by a twilight state of sleep and wakefulness.



Heroin

What is its effect on the body?

One of the most significant effects of heroin use is addiction. With regular heroin use, tolerance to the drug develops. Once this happens, the person must use more heroin to achieve the same intensity. As higher doses of the drug are used over time, physical dependence and addiction to the drug develop.

Effects of heroin use include:

- Drowsiness, respiratory depression, constricted pupils, nausea, a warm flushing of the skin, dry mouth, and heavy extremities

What are its overdose effects?

Because heroin users do not know the actual strength of the drug or its true contents, they are at a high risk of overdose or death.

The effects of a heroin overdose are:

- Slow and shallow breathing, blue lips and fingernails, clammy skin, convulsions, coma, and possible death

Which drugs cause similar effects?

Other opioids such as OxyContin®, Vicodin®, codeine, morphine, methadone, and fentanyl can cause similar effects as heroin.

What is its legal status in the United States?

Heroin is a Schedule I substance under the Controlled Substances Act meaning that it has a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.

Hydromorphone

WHAT IS HYDROMORPHONE?

Hydromorphone belongs to a class of drugs called "opioids," which includes morphine. It has an analgesic potency of two to eight times greater than that of morphine and has a rapid onset of action.

WHAT IS ITS ORIGIN?

Hydromorphone is legally manufactured and distributed in the United States. However, users can obtain hydromorphone from forged prescriptions, "doctor-shopping," theft from pharmacies, and from friends and acquaintances.

What are the street names?

Common street names include:

- D, Dillies, Dust, Footballs, Juice, and Smack

What does it look like?

Hydromorphone comes in:

- Tablets, capsules, oral solutions, and injectable formulations

How is it abused?

Users may abuse hydromorphone tablets by ingesting them. Injectable solutions, as well as tablets that have been crushed and dissolved in a solution may be injected as a substitute for heroin.

What is its effect on the mind?

When used as a drug of abuse, and not under a doctor's supervision, hydromorphone is taken to produce feelings of euphoria, relaxation, sedation, and reduced anxiety. It may also cause mental clouding, changes in mood, nervousness, and restlessness. It works centrally (in the brain) to reduce pain and suppress cough. Hydromorphone use is associated with both physiological and psychological dependence.

What is its effect on the body?

Hydromorphone may cause:

- Constipation, pupillary constriction, urinary retention, nausea, vomiting, respiratory depression, dizziness, impaired coordination, loss of appetite, rash, slow or rapid heartbeat, and changes in blood pressure

What are its overdose effects?

Acute overdose of hydromorphone can produce:

- Severe respiratory depression, drowsiness progressing to stupor or coma, lack of skeletal muscle tone, cold and clammy skin, constricted pupils, and reduction in blood pressure and heart rate

Severe overdose may result in death due to respiratory depression.

Which drugs cause similar effects?

Drugs that have similar effects include:

- Heroin, morphine, hydrocodone, fentanyl, and oxycodone

What is its legal status in the United States?

Hydromorphone is a Schedule II drug under the Controlled Substances Act with an accepted medical use as a pain reliever. Hydromorphone has a high potential for abuse and use may lead to severe psychological or physical dependence.

Methadone

WHAT IS METHADONE?

Methadone is a synthetic (man-made) narcotic.

WHAT IS ITS ORIGIN?

German scientists synthesized methadone during World War II because of a shortage of morphine. Methadone was introduced into the United States in 1947 as an analgesic (Dolophinel).

What are common street names?

Common street names include:

- Amidone, Chocolate Chip Cookies, Fizzies, Maria, Pastora, Salvia, Street Methadone, and Wafer

What does it look like?

Methadone is available as a tablet, oral solution, or injectable liquid. Tablets are available in 5 mg and 10 mg formulations. As of January 1, 2008, manufacturers of methadone hydrochloride tablets 40 mg (dispersible) have voluntarily agreed to restrict distribution of this formulation to only those facilities authorized for detoxification and maintenance treatment of opioid addiction, and hospitals. Manufacturers will instruct their wholesale distributors to discontinue supplying this formulation to any facility not meeting the above criteria.

How is it abused?

Methodone can be swallowed or injected.

What is its effect on the mind?

Abuse of methadone can lead to psychological dependence.

What is its effect on the body?

When an individual uses methadone, he/she may experience physical symptoms like sweating, itchy skin, or sleepiness. Individuals who abuse methadone risk becoming tolerant of and physically dependent on the drug.

When use is stopped, individuals may experience withdrawal symptoms including:

- Anxiety, muscle tremors, nausea, diarrhea, vomiting, and abdominal cramps

What are its overdose effects?

The effects of a methadone overdose are:

- Slow and shallow breathing, blue fingernails and lips, stomach spasms, clammy skin, convulsions, weak pulse, coma, and possible death

Which drugs cause similar effects?

Although chemically unlike morphine or heroin, methadone produces many of the same effects.

What is its legal status in the United States?

Methadone is a Schedule II drug under the Controlled Substances Act. While it may legally be used under a doctor's supervision, its non-medical use is illegal.



Methadone

Morphine

WHAT IS MORPHINE?

Morphine is a non-synthetic narcotic with a high potential for abuse and is derived from opium. It is used for the treatment of pain.

WHAT IS ITS ORIGIN?

In the United States, a small percentage of the morphine obtained from opium is used directly for pharmaceutical products. The remaining morphine is processed into codeine and other derivatives.

What are common street names?

Common street names include:

- Dreamer, Emsel, First Line, God's Drug, Hows, M.S., Mister Blue, Morf, Morpho, and Unkie

What does it look like?

Morphine is marketed under generic and brand name products, including:

- MS-Contin, Oramorph SR, MSIR, Roxanol, Kadian, and RMS

How is it abused?

Traditionally, morphine was almost exclusively used by injection, but the variety of pharmaceutical forms that it is marketed as today support its use by oral and other routes of administration.

Forms include:

- Oral solutions, immediate-and extended-release tablets and capsules, and injectable preparations

Those dependent on morphine prefer injection because the drug enters the bloodstream more quickly.

What is its effect on the mind?

Morphine's effects include euphoria and relief of pain. Chronic use of morphine results in tolerance and physical and psychological dependence.

What is its effect on the body?

Morphine use results in relief from physical pain, decrease in hunger, and inhibition of the cough reflex.

What are its overdose effects?

Overdose effects include:

- Cold and clammy skin, lowered blood pressure, sleepiness, slowed breathing, slow pulse rate, coma, and possible death

Which drugs cause similar effects?

Drugs causing similar effects as morphine include:

- Opium, codeine, heroin, methadone, hydrocodone, fentanyl, and oxycodone

What is its legal status in the United States?

Morphine is a Schedule II narcotic under the Controlled Substances Act.

Opium

WHAT IS OPIUM?

Opium is a highly addictive non-synthetic narcotic that is extracted from the poppy plant, Papaver somniferum. The opium poppy is the key source for many narcotics, including morphine, codeine, and heroin.

WHAT IS ITS ORIGIN?

The poppy plant, Papaver somniferum, is the source of opium. It was grown in the Mediterranean region as early as 5000 B.C., and has since been cultivated in a number of countries throughout the world. The milky fluid that seeps from its incisions in the unripe seed pod of this poppy has been scraped by hand and air-dried to produce what is known as opium.

A more modern method of harvesting for pharmaceutical use is by the industrial poppy straw process of extracting alkaloids from the mature dried plant (concentrate of poppy straw). All opium and poppy straw used for pharmaceutical products are imported into the United States from legitimate sources in regulated countries.

What are common street names?

Common street names include:

- Ah-pen-yen, Aunti, Aunti Emma, Big O, Black Pill, Chandoo, Chandu, Chinese Molasses, Chinese Tobacco, Dopium, Dover's Powder, Dream Gun, Dream Stick, Dreams, Easing Powder, Fi-do-nie, Gee, God's Medicine, Gondola, Goric, Great Tobacco, Guma, Hop/hops, Joy Plant, Midnight Oil, Mira, O, O.P., Ope, Pen Yan, Pin Gon, Pox, Skee, Toxy, Toys, When-shee, Ze, and Zero

What does it look like?

Opium can be a liquid, solid, or powder, but most poppy straw concentrate is available commercially as a fine brownish powder.

How is it abused?

Opium can be smoked, intravenously injected, or taken in pill form. Opium is also abused in combination with other drugs. For example, "Black" is a combination of marijuana, opium, and methamphetamine, and "Buddha" is potent marijuana spiked with opium.

What is its effect on the mind?

The intensity of opium's euphoric effects on the brain depends on the dose and route of administration. It works quickly when smoked because the opiate chemicals pass into the lungs, where they are quickly absorbed and then sent to the brain. An opium "high" is very similar to a heroin "high"; users experience a euphoric rush, followed by relaxation and the relief of physical pain.

What is its effect on the body?

Opium inhibits muscle movement in the bowels leading to constipation. It also can dry out the mouth and mucous membranes in the nose. Opium use leads to physical and psychological dependence, and can lead to overdose.

What are its overdose effects?

Overdose effects include:

- Slow breathing, seizures, dizziness, weakness, loss of consciousness, coma, and possible death

Which drugs cause similar effects?

Drugs that cause similar effects include:

- Morphine, codeine, heroin, methadone, hydroquinone, fentanyl, and oxycodone

What is its legal status in the United States?

Opium is a Schedule II drug under the Controlled Substances Act. Most opioids are Schedule II, III, IV, or V drugs. Some drugs that are derived from opium, such as heroin, are Schedule I drugs.

Oxycodone

WHAT IS OXYCODONE?

Oxycodone is a semi-synthetic narcotic analgesic and historically has been a popular drug of abuse among the narcotic abusing population.

WHAT IS ITS ORIGIN?

Oxycodone is synthesized from thebaine, a constituent of the poppy plants.

What are common street names?

Common street names include:

- Hillbilly Heroin, Kicker, OC, Ox, Roxy, Perc, and Oxy

What does it look like?

Oxycodone is marketed alone as OxyContin in 10, 20, 40 and 80 mg extended-release tablets and other immediate-release capsules like 5 mg OxyIR. It is also marketed in combination products with aspirin such as Percodan or acetaminophen such as Roxicet.

How is it abused?

Oxycodone is abused orally or intravenously. The tablets are crushed and sniffed or dissolved in water and injected. Others heat a tablet that has been placed on a piece of foil then inhale the vapors.

What is its effect on the mind?

Euphoria and feelings of relaxation are the most common effects of oxycodone on the brain, which explains its high potential for abuse.

What is its effect on the body?

Physiological effects of oxycodone include:

- Pain relief, sedation, respiratory depression, constipation, papillary constriction, and cough suppression. Extended or chronic use of oxycodone containing acetaminophen may cause severe liver damage

What are its overdose effects?

Overdose effects include:

- Extreme drowsiness, muscle weakness, confusion, cold and clammy skin, pinpoint pupils, shallow breathing, slow heart rate, fainting, coma, and possible death

Which drugs cause similar effects?

Drugs that cause similar effects to Oxycodone include:

- Opium, codeine, heroin, methadone, hydrocodone, fentanyl, and morphine

What is its legal status in the United States?

Oxycodone products are in Schedule II of the Controlled Substances Act.

V. Stimulants

WHAT ARE STIMULANTS?

- Stimulants speed up the body's systems. This class of drugs includes:
- Prescription drugs such as amphetamines [Adderall and dexedrine], methylphenidate [Concerta and Ritalin], diet aids [such as didrex, Bontril, Preludin, Fastin, Adipex P, ionomin, and Meridia] and illicitly produced drugs such as methamphetamine, cocaine, and methcathinone.

WHAT IS THEIR ORIGIN?

Stimulants are diverted from legitimate channels and clandestinely manufactured exclusively for the illicit market.



Ritalin SR 20mg tablet



Crack Cocaine

What are common street names?

Common street names for stimulants include:

- Bennies, Black Beauties, Cat, Coke, Crank, Crystal, Flake, Ice, Pellets, R-Ball, Skippy, Snow, Speed, Uppers, and Vitamin R

What do they look like?

Stimulants come in the form of:

- Pills, powder, rocks, and injectable liquids

How are they abused?

Stimulants can be pills or capsules that are swallowed. Smoking, snorting, or injecting stimulants produces a sudden sensation known as a "rush" or a "flash."

Abuse is often associated with a pattern of binge use — sporadically consuming large doses of stimulants over a short period of time. Heavy users may inject themselves every few hours, continuing until they have depleted their drug supply or reached a point of delirium, psychosis, and physical exhaustion. During heavy use, all other interests become secondary to recreating the initial euphoric rush.

What is their effect on the mind?

When used as drugs of abuse and not under a doctor's supervision, stimulants are frequently taken to:

- Produce a sense of exhilaration, enhance self-esteem, improve mental and physical performance, increase activity, reduce appetite, extend wakefulness for prolonged period, and "get high"

Chronic, high-dose use is frequently associated with agitation, hostility, panic, aggression, and suicidal or homicidal tendencies.

Paranoia, sometimes accompanied by both auditory and visual hallucinations, may also occur.

Tolerance, in which more and more drug is needed to produce the usual effects, can develop rapidly, and psychological dependence occurs. In fact, the strongest psychological dependence observed occurs with the more potent stimulants,

such as amphetamine, methylphenidate, methamphetamine, cocaine, and methcathinone.

Abrupt cessation is commonly followed by depression, anxiety, drug craving, and extreme fatigue, known as a "crash."

What is their effect on the body?

Stimulants are sometimes referred to as uppers and reverse the effects of fatigue on both mental and physical tasks. Therapeutic levels of stimulants can produce exhilaration, extended wakefulness, and loss of appetite. These effects are greatly intensified when large doses of stimulants are taken.

Taking too large a dose at one time or taking large doses over an extended period of time may cause such physical side effects as:

- Dizziness, tremors, headache, flushed skin, chest pain with palpitations, excessive sweating, vomiting, and abdominal cramps.

What are their overdose effects?

In overdose, unless there is medical intervention, high fever, convulsions, and cardiovascular collapse may precede death. Because accidental death is partially due to the effects of stimulants on the body's cardiovascular and temperature-regulating systems, physical exertion increases the hazards of stimulant use.

Which drugs cause similar effects?

Some hallucinogenic substances, such as ecstasy, have a stimulant component to their activity.

What is their legal status in the United States?

A number of stimulants have no medical use in the United States but have a high potential for abuse. These stimulants are controlled in Schedule I. Some prescription stimulants are not controlled, and some stimulants like tobacco and caffeine don't require a prescription — though society's recognition of their adverse effects has resulted in a proliferation of caffeine-free products and efforts to discourage cigarette smoking.

Stimulant chemicals in over-the-counter products, such as ephedrine and pseudoephedrine, can be found in allergy and cold medicine. As required by The Combat Methamphetamine Epidemic Act of 2005, a retail outlet must store these products out of reach of customers, either behind the counter or in a locked cabinet. Regulated sellers are required to maintain a written or electronic form of a logbook to record sales of these products. In order to purchase these products, customers must now show a photo identification issued by a state or federal government. They are also required to write or enter into the logbook: their name, signature, address, date, and time of sale. In addition to the above, there are daily and monthly sales limits set for customers.

Amphetamines

WHAT ARE AMPHETAMINES?

Amphetamines are stimulants that speed up the body's system. Many are legally prescribed and used to treat attention-deficit hyperactivity disorder (ADHD).

WHAT IS THEIR ORIGIN?

Amphetamine was first marketed in the 1930s as Benzedrine in an over-the-counter inhaler to treat nasal congestion. By 1937 amphetamine was available by prescription in tablet form and was used in the treatment of the sleeping disorder narcolepsy and ADHD.

Over the years, the use and abuse of clandestinely produced amphetamines have spread. Today, clandestine laboratory production of amphetamines has mushroomed, and the abuse of the drug has increased dramatically.

What are common street names?

Common street names include:

- Bennies, Black Beauties, Crank, Ice, Speed, and Uppers

What do they look like?

Amphetamines can look like pills or powder. Common prescription amphetamines include methylphenidate (Ritalin or Ritalin SR), amphetamine and dextroamphetamine (Adderall), and dextroamphetamine (Dexedrine).

How are they abused?

Amphetamines are generally taken orally or injected. However, the addition of "ice," the slang name of crystallized methamphetamine hydrochloride, has promoted smoking as another mode of administration. Just as "crack" is smokable cocaine, "ice" is smokable methamphetamine.

What is their effect on the mind?

The effects of amphetamines and methamphetamine are similar to cocaine, but their onset is slower and their duration is longer. In contrast to cocaine, which is quickly removed from the brain and is almost completely metabolized, methamphetamine remains in the central nervous system longer, and a larger percentage of the drug remains unchanged in the body, producing prolonged stimulant effects.

Chronic abuse produces a psychosis that resembles schizophrenia and is characterized by paranoia, picking at the skin, preoccupation with one's own thoughts, and auditory and visual hallucinations. Violent and erratic behavior is frequently seen among chronic users of amphetamines and methamphetamine.

What is their effect on the body?

Physical effects of amphetamine use include:

- Increased blood pressure and pulse rates, insomnia, loss of appetite, and physical exhaustion

What are their overdose effects?

Overdose effects include:

- Agitation, increased body temperature, hallucinations, convulsions, and possible death

Which drugs cause similar effects?

Drugs that cause similar effects include:

- Dexmethylphendiate, phentermine, benzphetamine, phendimetrazine, cocaine, crack, methamphetamine, and khat

What is their legal status in the United States?

Amphetamines are Schedule II stimulants, which means that they have a high potential for abuse and a currently acceptable medical use (in FDA-approved products). Pharmaceutical products are available only through a prescription that cannot be refilled.

Cocaine

WHAT IS COCAINE?

Cocaine is an intense, euphoria-producing stimulant drug with strong addictive potential.

WHAT IS ITS ORIGIN?

Cocaine is derived from coca leaves grown in Bolivia, Peru, and Colombia. The cocaine manufacturing process takes place in remote jungle labs where the raw product undergoes a series of chemical transformations. Colombia produces about 90 percent of the cocaine powder reaching the United States. Most of the cocaine entering the United States comes through Mexico.

What are common street names?

Common street names include:

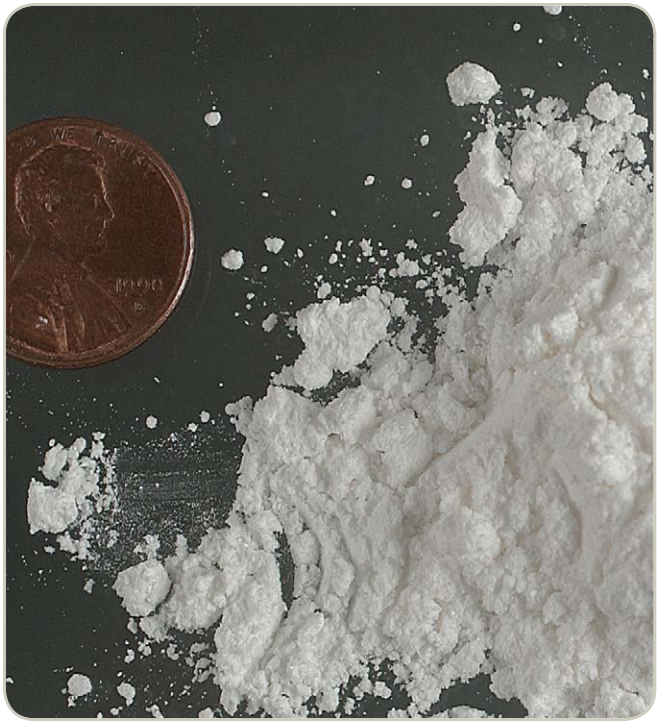
- Coca, Coke, Crack, Flake, Snow, and Soda Cot

What does it look like?

Cocaine is usually distributed as a white, crystalline powder. Cocaine is often diluted ("cut") with a variety of substances, the most common of which are sugars and local anesthetics. It is "cut" to stretch the amount of the product and increase profits for dealers. In contrast, cocaine base (crack) looks like small, irregularly shaped chunks (or "rocks") of a whitish solid.

How is it abused?

Powdered cocaine can be snorted or injected into the veins after dissolving in water. Cocaine base (crack) is smoked, either alone or on marijuana or tobacco. Cocaine is also used in combination with an opiate, like heroin, a practice known as "speedballing." Although injecting into veins or muscles, snorting, and smoking are the common ways of using cocaine, all mucous membranes readily absorb cocaine. Cocaine users typically binge on the drug until they are exhausted or run out of cocaine.



Cocaine powder

What is its effect on the mind?

The intensity of cocaine's euphoric effects depends on how quickly the drug reaches the brain, which depends on the dose and method of abuse. Following smoking or intravenous injection, cocaine reaches the brain in seconds, with a rapid buildup in levels. This results in a rapid-onset, intense euphoric effect known as a "rush."

By contrast, the euphoria caused by snorting cocaine is less intense and does not happen as quickly due to the slower build-up of the drug in the brain. Other effects include increased alertness and excitation, as well as restlessness, irritability, and anxiety.

Tolerance to cocaine's effects develops rapidly, causing users to take higher and higher doses. Taking high doses of cocaine or prolonged use, such as binging, usually causes paranoia. The crash that follows euphoria is characterized by mental and physical exhaustion, sleep, and depression lasting several days. Following the crash, users experience a craving to use cocaine again.

What is its effect on the body?

Physiological effects of cocaine include increased blood pressure and heart rate, dilated pupils, insomnia, and loss of appetite. The widespread abuse of highly pure street cocaine has led to many severe adverse health consequences such as:

- Cardiac arrhythmias, ischemic heart conditions, sudden cardiac arrest, convulsions, strokes, and death

In some users, the long-term use of inhaled cocaine has led to a unique respiratory syndrome, and chronic snorting of cocaine has led to the erosion of the upper nasal cavity.

Which drugs cause similar effects?

Other stimulants, such as methamphetamine, cause effects similar to cocaine that vary mainly in degree.

What is its legal status in the United States?

Cocaine is a Schedule II drug under the Controlled Substances Act, meaning it has a high potential for abuse and has an accepted medical use for treatment in the United States. Cocaine hydrochloride solution (4 percent and 10 percent) is used primarily as a topical local anesthetic for the upper respiratory tract. It also is used to reduce bleeding of the mucous membranes in the mouth, throat, and nasal cavities. However, better products have been developed for these purposes, and cocaine is rarely used medically in the United States.



Cocaine bricks, seized by DEA

Khat

WHAT IS KHAT?

Khat is a flowering evergreen shrub that is abused for its stimulant-like effect. Khat has two active ingredients, cathine and cathinone.

WHAT IS ITS ORIGIN?

Khat is native to East Africa and the Arabian Peninsula, where the use of it is an established cultural tradition for many social situations.



Khat plant

What are common street names?

Common street names for Khat include:

- Abyssinian Tea, African Salad, Catha, Chat, Kat, and Oat

What does it look like?

Khat is a flowering evergreen shrub. Khat that is sold and abused is usually just the leaves, twigs, and shoots of the Khat shrub.

How is it abused?

Khat is typically chewed like tobacco, then retained in the cheek and chewed intermittently to release the active drug, which produces a stimulant-like effect. Dried Khat leaves can be made into tea or a chewable paste, and Khat can also be smoked and even sprinkled on food.

What is its effect on the mind?

Khat can induce manic behavior with:

- Grandiose delusions, paranoia, nightmares, hallucinations, and hyperactivity

Chronic Khat abuse can result in violence and suicidal depression.

What is its effect on the body?

Khat causes an immediate increase in blood pressure and heart rate. Khat can also cause a brown staining of the teeth, insomnia, and gastric disorders. Chronic abuse of Khat can cause physical exhaustion.

What are its overdose effects?

The dose needed to constitute an overdose is not known, however it has been historically associated with those who are long-term chewers of the leaves. Symptoms of toxicity include:

- Delusions, loss of appetite, difficulty with breathing, and increases in both blood pressure and heart rate

Additionally, there are reports of liver damage (chemical hepatitis) and of cardiac complications, specifically myocardial infarctions. This mostly occurs among long-term chewers of khat or those who have chewed too large a dose.

Which drugs cause similar effects?

Khat's effects are similar to other stimulants, such as cocaine, amphetamine, and methamphetamine.

What is its legal status in the United States?

The chemicals found in khat are controlled under the Controlled Substances Act. Cathine is a Schedule IV stimulant, and cathinone is a Schedule I stimulant under the Controlled Substances Act, meaning that it has a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.

Methamphetamine

WHAT IS METHAMPHETAMINE?

Methamphetamine (meth) is a stimulant. The FDA-approved brand-name medication is Desoxyn.

WHAT IS ITS ORIGIN?

Mexican drug trafficking organizations have become the primary manufacturers and distributors of methamphetamine to cities throughout the United States, including in Hawaii. Domestic clandestine laboratory operators also produce and distribute meth but usually on a smaller scale. The methods used depend on the availability of precursor chemicals.

Currently, this domestic clandestinely produced meth is mainly made with diverted products that contain pseudoephedrine. Mexican methamphetamine is made with different precursor chemicals. The Combat Methamphetamine Epidemic Act of 2005 requires retailers of non-prescription products containing pseudoephedrine, ephedrine, or phenylpropanolamine to place these products behind the counter or in a locked cabinet. Consumers must show identification and sign a logbook for each purchase.

What are common street names?

Common street names include:

- Batu, Bikers Coffee, Black Beauties, Chalk, Chicken Feed, Crank, Crystal, Glass, Go-Fast, Hiropon, Ice, Meth, Methlies Quick, Poor Man's Cocaine, Shabu, Shards, Speed, Stove Top, Tina, Trash, Tweak, Uppers, Ventana, Vidrio, Yaba, and Yellow Bam

What does it look like?

Regular meth is a pill or powder. Crystal meth resembles glass fragments or shiny blue-white "rocks" of various sizes.

How is it abused?

Meth is swallowed, snorted, injected, or smoked. To intensify the effects, users may take higher doses of the drug, take it more frequently, or change their method of intake.



Methamphetamine in finished form

What is its effect on the mind?

Meth is a highly addictive drug with potent central nervous system (CNS) stimulant properties.

Those who smoke or inject it report a brief, intense sensation, or rush. Oral ingestion or snorting produces a long-lasting high instead of a rush, which reportedly can continue for as long as half a day. Both the rush and the high are believed to result from the release of very high levels of the neurotransmitter dopamine into areas of the brain that regulate feelings of pleasure. Long-term meth use results in many damaging effects, including addiction.

Chronic meth users can exhibit violent behavior, anxiety, confusion, insomnia, and psychotic features including paranoia, aggression, visual and auditory hallucinations, mood disturbances, and delusions — such as the sensation of insects creeping on or under the skin.



Methamphetamine in finished form

Such paranoia can result in homicidal or suicidal thoughts. Researchers have reported that as much as 50 percent of the dopamine-producing cells in the brain can be damaged after prolonged exposure to relatively low levels of meth. Researchers also have found that serotonin-containing nerve cells may be damaged even more extensively.

What is its effect on the body?

Taking even small amounts of meth can result in:

- Increased wakefulness, increased physical activity, decreased appetite, rapid breathing and heart rate, irregular heartbeat, increased blood pressure, and hyperthermia (overheating)

High doses can elevate body temperature to dangerous, sometimes lethal, levels, and cause convulsions and even cardiovascular collapse and death. Meth use may also cause extreme anorexia, memory loss, and severe dental problems.

What are its overdose effects?

High doses may result in death from stroke, heart attack, or multiple organ problems caused by overheating.

Which drugs cause similar effects?

Cocaine and potent stimulant pharmaceuticals, such as amphetamines and methylphenidate, produce similar effects.

What is its legal status in the United States?

Methamphetamine is a Schedule II stimulant under the Controlled Substances Act, which means that it has a high potential for abuse and a currently accepted medical use (in FDA-approved products). It is available only through a prescription that cannot be refilled. Today there is only one legal meth product, Desoxyn. It is currently marketed in 5-milligram tablets and has very limited use in the treatment of obesity and attention deficit hyperactivity disorder (ADHD).

VI. Depressants

WHAT ARE DEPRESSANTS?

Depressants will put you to sleep, relieve anxiety and muscle spasms, and prevent seizures.

Barbiturates are older drugs and include butalbital (Fiorina), phenobarbital, Pentothal, Seconal, and Nembutal. A person can rapidly develop dependence on and tolerance to barbiturates, meaning a person needs more and more of them to feel and function normally. This makes them unsafe, increasing the likelihood of coma or death.

Benzodiazepines were developed to replace barbiturates, though they still share many of the undesirable side effects including tolerance and dependence. Some examples are Valium, Xanax, Halcion, Ativan, Klonopin, and Restoril. Rohypnol is a benzodiazepine that is not manufactured or legally marketed in the United States, but it is used illegally.

Lunesta, Ambien, and Sonata are sedative-hypnotic medications approved for the short-term treatment of insomnia that share many of the properties of benzodiazepines. Other CNS depressants include meprobamate, methaqualone (Quaalude), and the illicit drug GHB.

WHAT IS THEIR ORIGIN?

Generally, legitimate pharmaceutical products are diverted to the illicit market. Teens can obtain depressants from the family medicine cabinet, friends, family members, the Internet, doctors, and hospitals.



Klonopin 5mg tablet



Blister pack of Rohypnol tablets

What are common street names?

Common street names for depressants include:

- Barbs, Benzos, Downers, Georgia Home Boy, GHB, Grievous Bodily Harm, Liquid X, Nerve Pills, Phennies, R2, Reds, Roofies, Rophies, Tranks, and Yellows

What do they look like?

Depressants come in the form of pills, syrups, and injectable liquids.

How are they abused?

Individuals abuse depressants to experience euphoria. Depressants are also used with other drugs to add to the other drugs' high or to deal with their side effects. Users take higher doses than people taking the drugs under a doctor's supervision for therapeutic purposes. Depressants like GHB and Rohypnol are also misused to facilitate sexual assault.

What is their effect on the mind?

Depressants used therapeutically do what they are prescribed for:

- To induce sleep, relieve anxiety and muscle spasms, and prevent seizures

They also:

- Cause amnesia, leaving no memory of events that occur while under the influence, reduce reaction time, impair mental functioning and judgment, and cause confusion

Long-term use of depressants produces psychological dependence and tolerance.

What is their effect on the body?

Some depressants can relax the muscles. Unwanted physical effects include:

- Slurred speech, loss of motor coordination, weakness, headache, lightheadedness, blurred vision, dizziness, nausea, vomiting, low blood pressure, and slowed breathing

Prolonged use of depressants can lead to physical dependence even at doses recommended for medical treatment. Unlike barbiturates, large doses of benzodiazepines are rarely fatal unless combined with other drugs or alcohol. But unlike the withdrawal syndrome seen with most other drugs of abuse, withdrawal from depressants can be life threatening.



Vials containing GHB

What is their legal status in the United States?

Most depressants are controlled substances that range from Schedule I to Schedule IV under the Controlled Substances Act, depending on their risk for abuse and whether they currently have an accepted medical use. Many of the depressants have FDA-approved medical uses. Rohypnol and Quaaludes are not manufactured or legally marketed in the United States.

Barbiturates

WHAT ARE BARBITURATES?

Barbiturates are depressants that produce a wide spectrum of central nervous system depression from mild sedation to coma. They also have been used as sedatives, hypnotics, anesthetics, and anticonvulsants.

Barbiturates are classified as:

- Ultrashort, Short, Intermediate, Long-acting

WHAT IS THEIR ORIGIN?

Barbiturates were first introduced for medical use in the 1900s, and today about 12 substances are in medical use.

What are common street names?

Common street names include:

- Barbs, Block Busters, Christmas Trees, Goof Balls, Pinks, Red Devils, Reds & Blues, and Yellow Jackets

What do they look like?

Barbiturates come in a variety of multicolored pills and tablets. Users prefer the short-acting and intermediate barbiturates such as Amytal and Seconal.

How are they abused?

Barbiturates are abused by swallowing a pill or injecting a liquid form. Barbiturates are generally abused to reduce anxiety, decrease inhibitions, and treat unwanted effects of illicit drugs. Barbiturates can be extremely dangerous because overdoses can occur easily and lead to death.

What is their effect on the mind?

Barbiturates cause:

- Mild euphoria, lack of inhibition, relief of anxiety, and sleepiness

Higher doses cause:

- Impairment of memory, judgment, and coordination; irritability; and paranoid and suicidal ideation

Tolerance develops quickly and larger doses are then needed to produce the same effect, increasing the danger of an overdose.

What is their effect on the body?

Barbiturates slow down the central nervous system and cause sleepiness.

What are their overdose effects?

Effects of overdose include:

- Shallow respiration, clammy skin, dilated pupils, weak and rapid pulse, coma, and possible death

Which drugs cause similar effects?

Drugs with similar effects include:

- Alcohol, benzodiazepines like Valium and Xanax, tranquilizers, sleeping pills, Rohypnol, and GHB

What is their legal status in the United States?

Barbiturates are Schedule II, III, and IV depressants under the Controlled Substances Act.

Benzodiazepines

WHAT ARE BENZODIAZEPINES?

Benzodiazepines are depressants that produce sedation and hypnosis, relieve anxiety and muscle spasms, and reduce seizures.

WHAT IS THEIR ORIGIN?

Benzodiazepines are only legally available through prescription. Many users maintain their drug supply by getting prescriptions from several doctors, forging prescriptions, or buying them illicitly. Alprazolam and diazepam are the two most frequently encountered benzodiazepines on the illicit market.

What are common street names?

Common street names include Benzos and Downers.

What do they look like?

The most common benzodiazepines are the prescription drugs Valium, Xanax, Halcion, Ativan, and Klonopin. Tolerance can develop, although at variable rates and to different degrees. Shorter-acting benzodiazepines used to manage insomnia include estazolam (ProSom), flurazepam (Dalmane), temazepam (Restoril), and triazolam (Halcion). Midazolam (Versed), a short-acting benzodiazepine, is utilized for sedation, anxiety, and amnesia in critical care settings and prior to anesthesia. It is available in the United States as an injectable preparation and as a syrup (primarily for pediatric patients).

Benzodiazepines with a longer duration of action are utilized to treat insomnia in patients with daytime anxiety. These benzodiazepines include alprazolam (Xanax), chlordiazepoxide (Librium), clorazepate (Tranxene), diazepam (Valium), halazepam (Paxipam), lorzepam (Ativan), oxazepam (Serax), prazepam (Centrax), and quazepam (Doral). Clonazepam (Klonopin), diazepam, and clorazepate are also used as anticonvulsants.

How are they abused?

Abuse is frequently associated with adolescents and young adults who take the drug orally or crush it up and snort it to get high. Abuse is particularly high among heroin and cocaine users.

What is their effect on the mind?

Benzodiazepines are associated with amnesia, hostility, irritability, and vivid or disturbing dreams.

What is their effect on the body?

Benzodiazepines slow down the central nervous system and may cause sleepiness.

What are their overdose effects?

Effects of overdose include:

- Shallow respiration, clammy skin, dilated pupils, weak and rapid pulse, coma, and possible death

Which drugs cause similar effects?

Drugs that cause similar effects include:

- Alcohol, barbiturates, sleeping pills, and GHB

What is their legal status in the United States?

Benzodiazepines are controlled in Schedule IV of the Controlled Substances Act.

GHB

WHAT IS GHB?

Gamma-Hydroxybutyric acid (GHB) is another name for the generic drug sodium oxybate. Xyrem (which is sodium oxybate) is the trade name of the Food and Drug Administration (FDA)-approved prescription medication.

Analogues that are often substituted for GHB include GBL (gamma butyrolactone) and 1,4 BD (also called just "BD"), which is 1,4-butanediol. These analogues are available legally as industrial solvents used to produce polyurethane, pesticides, elastic fibers, pharmaceuticals, coatings on metal or plastic, and other products. They are also sold illicitly as supplements for bodybuilding, fat loss, reversal of baldness, improved eyesight, and to combat aging, depression, drug addiction, and insomnia.

GBL and BD are sold as "fish tank cleaner," "ink stain remover," "ink cartridge cleaner," and "nail enamel remover" for approximately $100 per bottle — much more expensive than comparable products. Attempts to identify the abuse of GHB analogues are hampered by the fact that routine toxicological screens do not detect the presence of these analogues.

WHAT IS ITS ORIGIN?

GHB is produced illegally in both domestic and foreign clandestine laboratories. The major source of GHB on the street is through clandestine synthesis by local operators. At bars or "rave" parties, GHB is typically sold in liquid form by the capful or "swig" for $5 to $25 per cap. Xyrem has the potential for diversion and abuse like any other pharmaceutical containing a controlled substance.

GHB has been encountered in nearly every region of the country.

What are common street names?

Common street names include:

- Easy Lay, G, Georgia Home Boy, GHB, Goop, Grievous Bodily Harm, Liquid Ecstasy, Liquid X, and Scoop



Vials containing GHB

What does it look like?

GHB is usually sold as a liquid or as a white powder that is dissolved in a liquid, such as water, juice, or alcohol. GHB dissolved in liquid has been packaged in small vials or small water bottles. In liquid form, GHB is clear and colorless and slightly salty in taste.

How is it abused?

GHB and its analogues are abused for their euphoric and calming effects and because some people believe they build muscles and cause weight loss.

GHB and its analogues are also misused for their ability to increase libido, suggestibility, passivity, and to cause amnesia (no memory of events while under the influence of the substance) — traits that make users vulnerable to sexual assault and other criminal acts.

GHB abuse became popular among teens and young adults at dance clubs and "raves" in the 1990s and gained notoriety as a date rape drug. GHB is taken alone or in combination with other drugs, such as alcohol (primarily), other depressants, stimulants, hallucinogens, and marijuana.

The average dose ranges from 1 to 5 grams (depending on the purity of the compound, this can be 1-2 teaspoons mixed in a beverage). However, the concentrations of these "home-brews" have varied so much that users are usually unaware of the actual dose they are drinking.

What is its effect on the mind?

GHB occurs naturally in the central nervous system in very small amounts. Use of GHB produces Central Nervous System (CNS) depressant effects including:

- Euphoria, drowsiness, decreased anxiety, confusion, and memory impairment

GHB can also produce both visual hallucinations and — paradoxically — excited and aggressive behavior. GHB greatly increases the CNS depressant effects of alcohol and other depressants.

What is its effect on the body?

GHB takes effect in 15 to 30 minutes, and the effects last 3 to 6 hours. Low doses of GHB produce nausea.

At high doses, GHB overdose can result in:

- Unconsciousness, seizures, slowed heart rate, greatly slowed breathing, lower body temperature, vomiting, nausea, coma, and death

Regular use of GHB can lead to addiction and withdrawal that includes:

- Insomnia, anxiety, tremors, increased heart rate and blood pressure, and occasional psychotic thoughts

Currently, there is no antidote available for GHB intoxication. GHB analogues are known to produce side effects such as:

- Topical irritation to the skin and eyes, nausea, vomiting, incontinence, loss of consciousness, seizures, liver damage, kidney failure, respiratory depression, and death

What are its overdose effects?

GHB overdose can cause death.

Which drugs cause similar effects?

GHB analogues are often abused in place of GHB. Both GBL and BD metabolize to GHB when taken and produce effects similar to GHB.

CNS depressants such as barbiturates and methaqualone also produce effects similar to GHB.

What is its legal status in the United States?

GHB is a Schedule I controlled substance, meaning that it has a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision. FDA-approved GHB products are Schedule III substances under the Controlled Substances Act. In addition, GBL is a List I chemical.

It was placed on Schedule I of the Controlled Substances Act in March 2000. However, when sold as FDA-approved GHB products (such as Xyrem), it is considered Schedule III, one of several drugs that are listed in multiple schedules.

Rohypnol

WHAT IS ROHYPNOL?

Rohypnol is a trade name for flunitrazepam, a central nervous system (CNS) depressant that belongs to a class of drugs known as benzodiazepines. Flunitrazepam is also marketed as generic preparations and other trade name products outside of the United States.

Like other benzodiazepines, Rohypnol produces sedative-hypnotic, anti-anxiety, and muscle relaxant effects. This drug has never been approved for medical use in the United States by the Food and Drug Administration. Outside the United States, Rohypnol is commonly prescribed to treat insomnia. Rohypnol is also referred to as a "date rape" drug.

WHAT IS ITS ORIGIN?

Rohypnol is smuggled into the United States from other countries, such as Mexico.

What are common street names?

Common street names include:

- Circles, Forget Pill, Forget-Me-Pill, La Rocha, Lunch Money Drug, Mexican Valium, Pingus, R2, Reynolds, Roach, Roach 2, Roaches, Roachies, Roapies, Robutal, Rochas Dos, Rohypnol, Roofies, Rophies, Ropies, Roples, Row-Shay, Ruffies, and Wolfies

What does it look like?

Prior to 1997, Rohypnol was manufactured as a white tablet (0.5-2 milligrams per tablet), and when mixed in drinks, was colorless, tasteless, and odorless. In 1997, the manufacturer responded to concerns about the drug's role in sexual assaults by reformulating the drug.

Rohypnol is now manufactured as an oblong olive green tablet with a speckled blue core that when dissolved in light-colored drinks will dye the liquid blue. However, generic versions of the drug may not contain the blue dye.

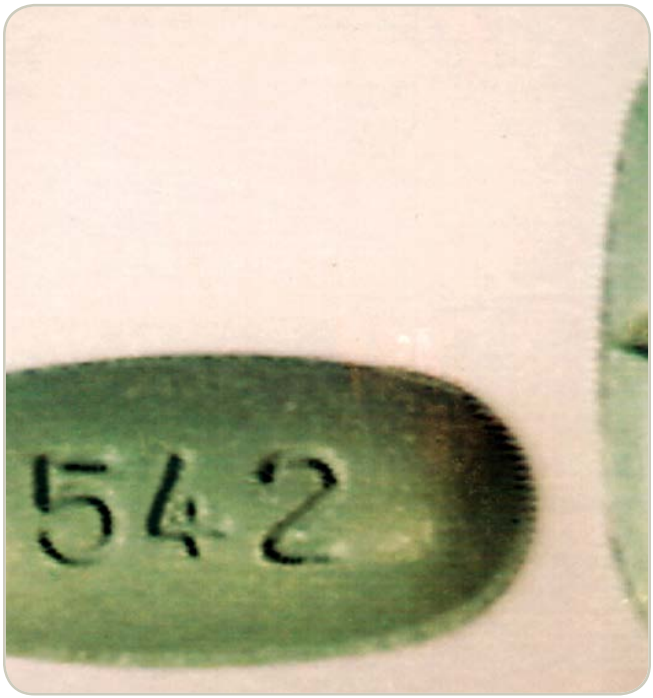
How is it abused?

The tablet can be swallowed whole, crushed and snorted, or dissolved in liquid. Adolescents may abuse Rohypnol to produce a euphoric effect often described as a "high." While high, they experience reduced inhibitions and impaired judgment.

Rohypnol is also used in combination with alcohol to produce an exaggerated intoxication.

In addition, abuse of Rohypnol may be associated with multiple-substance abuse. For example, cocaine users may use benzodiazepines such as Rohypnol to relieve the side effects (e.g., irritability and agitation) associated with cocaine binges.

Rohypnol is also misused to physically and psychologically incapacitate victims targeted for sexual assault. The drug is usually placed in the alcoholic drink of an unsuspecting victim to incapacitate them and prevent resistance to sexual assault. The drug leaves the victim unaware of what has happened to them.



Rohypnol tablets

What is its effect on the mind?

Like other benzodiazepines, Rohypnol slows down the functioning of the CNS producing:

- Drowsiness (sedation), sleep (pharmacological hypnosis), decreased anxiety, and amnesia (no memory of events while under the influence of the substance)

Rohypnol can also cause:

- Increased or decreased reaction time, impaired mental functioning and judgment, confusion, aggression, and excitability

What is its effect on the body?

Rohypnol causes muscle relaxation. Adverse physical effects include:

- Slurred speech, loss of motor coordination, weakness, headache, and respiratory depression

Rohypnol also can produce physical dependence when taken regularly over a period of time.

What are its overdose effects?

High doses of Rohypnol, particularly when combined with CNS depressant drugs such as alcohol and heroin, can cause severe sedation, unconsciousness, slow heart rate, and suppression of respiration that may be sufficient to result in death.

Which drugs cause similar effects?

Drugs that cause similar effects include GHB (gamma hydroxybutyrate) and other benzodiazepines such as alprazolam (e.g., Xanax), clonazepam (e.g., Klonopin), and diazepam (e.g., Valium).

What is its legal status in the United States?

Rohypnol is a Schedule IV substance under the Controlled Substances Act. Rohypnol is not approved for manufacture, sale, use, or importation to the United States. It is legally manufactured and marketed in many countries. Penalties for possession, trafficking, and distribution involving one gram or more are the same as those of a Schedule I drug.



Blister pack of Rohypnol tablets

VII. Hallucinogens

WHAT ARE HALLUCINOGENS?

Hallucinogens are found in plants and fungi or are synthetically produced and are among the oldest known group of drugs used for their ability to alter human perception and mood.

WHAT IS THEIR ORIGIN?

Hallucinogens can be synthetically produced in illicit laboratories or are found in plants.



MDMA/Ecstasy pills



LSD Blotter Sheet

What are common street names?

Common street names include:

- Acid, Blotter, Blotter Acid, Cubes, Doses, Fry, Mind Candy, Mushrooms, Shrooms, Special K, STP, X, and XTC

What do they look like?

Hallucinogens come in a variety of forms. MDMA or ecstasy tablets are sold in many colors with a variety of logos to attract youth. LSD is sold in the form of impregnated paper (blotter acid), typically imprinted with colorful graphic designs.

How are they abused?

The most commonly abused hallucinogens among junior and senior high school students are hallucinogenic mushrooms, LSD, and MDMA (ecstasy). Hallucinogens are typically taken orally or can be smoked.

What is their effect on the mind?

Sensory effects include perceptual distortions that vary with dose, setting, and mood. Psychic effects include distortions of thought associated with time and space. Time may appear to stand still, and forms and



LSD powder and capsules

colors seem to change and take on new significance. Weeks or even months after some hallucinogens have been taken, the user may experience flashbacks — fragmentary recurrences of certain aspects of the drug experience in the absence of actually taking the drug. The occurrence of a flashback is unpredictable, but is more likely to occur during times of stress and seems to occur more frequently in younger individuals. With time, these episodes diminish and become less intense.

What is their effect on the body?

Physiological effects include elevated heart rate, increased blood pressure, and dilated pupils.

What are their overdose effects?

Deaths exclusively from acute overdose of LSD, magic mushrooms, and mescaline are extremely rare. Deaths generally occur due to suicide, accidents, and dangerous behavior, or due to the person inadvertently eating poisonous plant material.

A severe overdose of PCP and ketamine can result in:

- Respiratory depression, coma, convulsions, seizures, and death due to respiratory arrest

What is their legal status in the United States?

Many hallucinogens are Schedule I under the Controlled Substances Act, meaning that they have a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.

Ecstasy/MDMA

WHAT IS ECSTASY/MDMA?

MDMA acts as both a stimulant and psychedelic, producing an energizing effect, distortions in time and perception, and enhanced enjoyment of tactile experiences.

Adolescents and young adults use it to reduce inhibitions and to promote:

- Euphoria, feelings of closeness, empathy, and sexuality

Although MDMA is known among users as ecstasy, researchers have determined that many ecstasy tablets contain not only MDMA but also a number of other drugs or drug combinations that can be harmful, such as:

- Methamphetamine, ketamine, cocaine, the over-the-counter cough suppressant dextromethorphan (DXM), the diet drug ephedrine, and caffeine

In addition, other drugs similar to MDMA, such as MDA or PMA, are often sold as ecstasy, which can lead to overdose and death when the user takes additional doses to obtain the desired effect.

WHAT IS ITS ORIGIN?

MDMA is a synthetic chemical made in labs. Seized MDMA in the U.S. is primarily manufactured in, and smuggled across our borders from, clandestine laboratories in Canada and, to a lesser extent, the Netherlands. A small number of MDMA clandestine laboratories have also been identified operating in the U.S.

What are common street names?

Common street names include:

- Adam, Beans, Clarity, Disco Biscuit, E, Ecstasy, Eve, Go, Hug Drug, Lover's Speed, MDMA, Peace, STP, X, and XTC

What does it look like?

MDMA is mainly distributed in tablet form. MDMA tablets are sold with logos, creating brand names for users to seek out. The colorful pills are often hidden among colorful candies. MDMA is also distributed in capsules, powder, and liquid forms.

How is it abused?

MDMA use mainly involves swallowing tablets (50-150 mg), which are sometimes crushed and snorted, occasionally smoked but rarely injected. MDMA is also available as a powder.

MDMA users usually take MDMA by "stacking" (taking three or more tablets at once) or by "piggy-backing" (taking a series of tablets over a short period of time). One trend among young adults is "candy flipping," which is the co-abuse of MDMA and LSD.

MDMA is considered a "party drug." As with many other drugs of abuse, MDMA is rarely used alone. It is common for users to mix MDMA with other substances, such as alcohol and marijuana.

What is its effect on the mind?

MDMA mainly affects brain cells that use the chemical serotonin to communicate with each other. Serotonin helps to regulate mood, aggression, sexual activity, sleep, and sensitivity to pain. Clinical studies suggest that MDMA may increase the risk of long-term, perhaps permanent, problems with memory and learning.

MDMA causes changes in perception, including euphoria and increased sensitivity to touch, energy, sensual and sexual arousal, need to be touched, and need for stimulation.

Some unwanted psychological effects include:

- Confusion, anxiety, depression, paranoia, sleep problems, and drug craving

All these effects usually occur within 30 to 45 minutes of swallowing the drug and usually last 4 to 6 hours, but they may occur or last weeks after ingestion.

What is its effect on the body? Users of MDMA experience many of the same effects and face many of the same risks as users of other stimulants such as cocaine and amphetamines. These include increased motor activity, alertness, heart rate, and blood pressure.



MDMA/Ecstasy pills

Some unwanted physical effects include:

- Muscle tension, tremors, involuntary teeth clenching, muscle cramps, nausea, faintness, chills, sweating, and blurred vision

High doses of MDMA can interfere with the ability to regulate body temperature, resulting in a sharp increase in body temperature (hyperthermia), leading to liver, kidney, and cardiovascular failure.

Severe dehydration can result from the combination of the drug's effects and the crowded and hot conditions in which the drug is often taken.

Studies suggest chronic use of MDMA can produce damage to the serotonin system. It is ironic that a drug that is taken to increase pleasure may cause damage that reduces a person's ability to feel pleasure.

What are its overdose effects?

In high doses, MDMA can interfere with the body's ability to regulate temperature. On occasions, this can lead to a sharp increase in body temperature (hyperthermia), resulting in liver, kidney, and cardiovascular system failure, and death. Because MDMA can interfere with its own metabolism (that is, its breakdown within the body), potentially harmful levels can be reached by repeated drug use within short intervals.

Which drugs cause similar effects?

MDMA produces both amphetamine-like stimulation and mild mescaline-like hallucinations.

What is its legal status in the United States?

MDMA is a Schedule I drug under the Controlled Substances Act, meaning it has a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.

Ketamine

WHAT IS KETAMINE?

Ketamine is a dissociative anesthetic that has some hallucinogenic effects. It distorts perceptions of sight and sound and makes the user feel disconnected and not in control. It is an injectable, short-acting anesthetic for use in humans and animals. It is referred to as a "dissociative anesthetic" because it makes patients feel detached from their pain and environment.

Ketamine can induce a state of sedation (feeling calm and relaxed), immobility, relief from pain, and amnesia (no memory of events while under the influence of the drug). It is abused for its ability to produce dissociative sensations and hallucinations. Ketamine has also been used to facilitate sexual assault.

WHAT IS ITS ORIGIN?

Ketamine is produced commercially in a number of countries, including the United States. Most of the ketamine illegally distributed in the United States is diverted or stolen from legitimate sources, particularly veterinary clinics, or smuggled into the United States from Mexico.

Distribution of ketamine typically occurs among friends and acquaintances, most often at raves, nightclubs, and at private parties; street sales of ketamine are rare.

What are common street names?

Common street names include:

- Cat Tranquilizer, Cat Valium, Jet K, Kit Kat, Purple, Special K, Special La Coke, Super Acid, Super K, and Vitamin K

What does it look like?

Ketamine comes in a clear liquid and a white or off-white powder. Powdered ketamine (100 milligrams to 200 milligrams) typically is packaged in small glass vials, small plastic bags, and capsules as well as paper, glassine, or aluminum foil folds.



Vials containing liquid ketamine

How is it abused?

Ketamine, along with the other "club drugs," has become popular among teens and young adults at dance clubs and "raves." Ketamine is manufactured commercially as a powder or liquid. Powdered ketamine is also formed from pharmaceutical ketamine by evaporating the liquid using hot plates, warming trays, or microwave ovens, a process that results in the formation of crystals, which are then ground into powder.

Powdered ketamine is cut into lines known as bumps and snorted, or it is smoked, typically in marijuana or tobacco cigarettes. Liquid ketamine is injected or mixed into drinks. Ketamine is found by itself or often in combination with MDMA, amphetamine, methamphetamine, or cocaine.

What is its effect on the mind?

Ketamine produces hallucinations. It distorts perceptions of sight and sound and makes the user feel disconnected and not in control. A "Special K" trip is touted as better than that of LSD or PCP because its hallucinatory effects are relatively short in duration, lasting approximately 30 to 60 minutes as opposed to several hours.

Slang for experiences related to Ketamine or effects of ketamine include:

- "K-land" (refers to a mellow & colorful experience)
- "K-hole" (refers to the out-of-body, near death experience)
- "Baby food" (users sink in to blissful, infantile inertia)
- "God" (users are convinced that they have met their maker)

The onset of effects is rapid and often occurs within a few minutes of taking the drug, though taking it orally results in a slightly slower onset of effects. Flashbacks have been reported several weeks after ketamine is used. Ketamine may also cause agitation, depression, cognitive difficulties, unconsciousness, and amnesia.

What is its effect on the body?

A couple of minutes after taking the drug, the user may experience an increase in heart rate and blood pressure that gradually decreases over the next 10 to 20 minutes. Ketamine can make users unresponsive to stimuli. When in this state, users experience:

- Involuntarily rapid eye movement, dilated pupils, salivation, tear secretions, and stiffening of the muscles

This drug can also cause nausea.

What are its overdose effects?

An overdose can cause unconsciousness and dangerously slowed breathing.

Which drugs cause similar effects?

Other hallucinogenic drugs such as LSD, PCP, and mescaline can cause hallucinations. There are also several drugs such as GHB, Rohypnol, and other depressants that are misused for their amnesiac or sedative properties to facilitate sexual assault.



Ketamine in various forms

What is its legal status in the United States?

Since the 1970s, ketamine has been marketed in the United States as an injectable, short-acting anesthetic for use in humans and animals. In 1999, ketamine including its salts, isomers and salts of isomers, became a Schedule III non-narcotic substance under the Controlled Substances Act. It has a currently accepted medical use but some potential for abuse, which may lead to moderate or low physical dependence or high psychological dependence.

LSD

WHAT IS LSD?

LSD is a potent hallucinogen that has a high potential for abuse and currently has no accepted medical use in treatment in the United States.

WHAT IS ITS ORIGIN?

LSD is produced in clandestine laboratories in the United States.

What are common street names?

Common names for LSD include:

- Acid, Blotter Acid, Dots, Mellow Yellow, and Window Pane

What does it look like?

LSD is sold on the street in tablets, capsules, and occasionally in liquid form. It is an odorless and colorless substance with a slightly bitter taste. LSD is often added to absorbent paper, such as blotter paper, and divided into small decorated squares, with each square representing one dose.

How is it abused?

LSD is abused orally.

What is its effect on the mind?

During the first hour after ingestion, users may experience visual changes with extreme changes in mood. While hallucinating, the user may suffer impaired depth and time perception accompanied by distorted perception of the shape and size of objects, movements, colors, sound, touch, and the user's own body image.

The ability to make sound judgments and see common dangers is impaired, making the user susceptible to personal injury. It is possible for users to suffer acute anxiety and depression after an LSD "trip" and flashbacks have been reported days, and even months, after taking the last dose.



LSD powder

What is its effect on the body?

The physical effects include:

- Dilated pupils, higher body temperature, increased heart rate and blood pressure, sweating, loss of appetite, sleeplessness, dry mouth, and tremors

What are its overdose effects?

Longer, more intense "trip" episodes, psychosis, and possible death

Which drugs cause similar effects?

LSD's effects are similar to other hallucinogens, such as PCP, mescaline, and peyote.

What is its legal status in the United States?

LSD is a Schedule I substance under the Controlled Substances Act. Schedule I substances have a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.

Peyote & Mescaline

WHAT ARE PEYOTE AND MESCALINE?

Peyote is a small, spineless cactus. The active ingredient in peyote is the hallucinogen mescaline.

WHAT IS ITS ORIGIN?

From earliest recorded time, peyote has been used by natives in northern Mexico and the southwestern United States as a part of their religious rites. Mescaline can be extracted from peyote or produced synthetically.

What are common street names?

Common street names include:

- Buttons, Cactus, Mesc, and Peyoto

What does it look like?

The top of the peyote cactus is referred to as the "crown" and consists of disc-shaped buttons that are cut off.

How is it abused?

The fresh or dried buttons are chewed or soaked in water to produce an intoxicating liquid. Peyote buttons may also be ground into a powder that can be placed inside gelatin capsules to be swallowed, or smoked with a leaf material such as cannabis or tobacco.

What is its effect on the mind?

Abuse of peyote and mescaline will cause varying degrees of:

- Illusions, hallucinations, altered perception of space and time, and altered body image

Users may also experience euphoria, which is sometimes followed by feelings of anxiety.

What is its effect on the body?

Following the consumption of peyote and mescaline, users may experience:

- Intense nausea, vomiting, dilation of the pupils, increased heart rate, increased blood pressure, a rise in body temperature that causes heavy perspiration, headaches, muscle weakness, and impaired motor coordination

Which drugs cause similar effects?

Other hallucinogens like LSD, psilocybin (mushrooms), and PCP

What is its legal status in the United States?

Peyote and mescaline are Schedule I substances under the Controlled Substances Act, meaning that they have a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.



Peyote cactus

Psilocybin

WHAT IS PSILOCYBIN?

Psilocybin is a chemical obtained from certain types of fresh or dried mushrooms.

WHAT IS ITS ORIGIN?

Psilocybin mushrooms are found in Mexico, Central America, and the United States.

What are common street names?

Common street names include:

- Magic Mushrooms, Mushrooms, and Shrooms

What does it look like?

Mushrooms containing psilocybin are available fresh or dried and have long, slender stems topped by caps with dark gills on the underside. Fresh mushrooms have white or whitish-gray stems; the caps are dark brown around the edges and light brown or white in the center. Dried mushrooms are usually rusty brown with isolated areas of off-white.

How is it abused?

Psilocybin mushrooms are ingested orally. They may also be brewed as a tea or added to other foods to mask their bitter flavor.

What is its effect on the mind?

The psychological consequences of psilocybin use include hallucinations and an inability to discern fantasy from reality. Panic reactions and psychosis also may occur, particularly if a user ingests a large dose.



Psilocybin mushrooms

What is its effect on the body?

The physical effects include:

- Nausea, vomiting, muscle weakness, and lack of coordination

What are its overdose effects?

Effects of overdose include:

- Longer, more intense "trip" episodes, psychosis, and possible death

Abuse of psilocybin mushrooms could also lead to poisoning if one of the many varieties of poisonous mushrooms is incorrectly identified as a psilocybin mushroom.

Which drugs cause similar effects?

Psilocybin effects are similar to other hallucinogens, such as mescaline and peyote.

What is its legal status in the United States?

Psilocybin is a Schedule I substance under the Controlled Substances Act, meaning that it has a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.



VIII. Marijuana/Cannabis

WHAT IS MARIJUANA?

Marijuana is a mind-altering (psychoactive) drug, produced by the Cannabis sativa plant. Marijuana contains over 480 constituents. THC (delta-9-tetrahydrocannabinol) is believed to be the main ingredient that produces the psychoactive effect.

WHAT IS ITS ORIGIN?

Marijuana is grown in the United States, Canada, Mexico, South America, Caribbean, and Asia. It can be cultivated in both outdoor and indoor settings.

What are common street names?

Common street names include:

- Aunt Mary, BC Bud, Blunts, Boom, Chronic, Dope, Gangster, Ganja, Grass, Hash, Herb, Hydro, Indo, Joint, Kif, Mary Jane, Mota, Pot, Reefer, Sinsemilla, Skunk, Smoke, Weed, and Yerba

What does it look like?

Marijuana is a dry, shredded green/brown mix of flowers, stems, seeds, and leaves from the Cannabis sativa plant. The mixture typically is green, brown, or gray in color and may resemble tobacco.

How is it abused?

Marijuana is usually smoked as a cigarette (called a joint) or in a pipe or bong. It is also smoked in blunts, which are cigars that have been emptied of tobacco and refilled with marijuana, sometimes in combination with another drug. Marijuana is also mixed with foods or brewed as a tea.

What is its effect on the mind?

When marijuana is smoked, the THC passes from the lungs and into the bloodstream, which carries the chemical to the organs throughout the body, including the brain. In the brain, the THC connects to specific sites called cannabinoid receptors on nerve cells and influences the activity of those cells.

Many of these receptors are found in the parts of the brain that influence:

- Pleasure, memory, thought, concentration, sensory and time perception, and coordinated movement

The short-term effects of marijuana include:

- Problems with memory and learning, distorted perception, difficulty in thinking and problem-solving, and loss of coordination

The effect of marijuana on perception and coordination are responsible for serious impairments in learning, associative processes, and psychomotor behavior (driving abilities). Long term, regular use can lead to physical dependence and withdrawal following discontinuation, as well as psychic addiction or dependence.

Clinical studies show that the physiological, psychological, and behavioral effects of marijuana vary among individuals and present a list of common responses to cannabinoids, as described in the scientific literature:

- Dizziness, nausea, tachycardia, facial flushing, dry mouth, and tremor initially
- Merriment, happiness, and even exhilaration at high doses
- Disinhibition, relaxation, increased sociability, and talkativeness
- Enhanced sensory perception, giving rise to increased appreciation of music, art, and touch

- Heightened imagination leading to a subjective sense of increased creativity
- Time distortions
- Illusions, delusions, and hallucinations are rare except at high doses
- Impaired judgment, reduced coordination, and ataxia, which can impede driving ability or lead to an increase in risk-taking behavior
- Emotional lability, incongruity of affect, dysphoria, disorganized thinking, inability to converse logically, agitation, paranoia, confusion, restlessness, anxiety, drowsiness, and panic attacks may occur, especially in inexperienced users or in those who have taken a large dose
- Increased appetite and short-term memory impairment are common

What is its effect on the body?

Short-term physical effects from marijuana use may include:

- Sedation, bloodshot eyes, increased heart rate, coughing from lung irritation, increased appetite, and decreased blood pressure

Marijuana smokers experience serious health problems such as bronchitis, emphysema, and bronchial asthma. Extended use may cause suppression of the immune system. Withdrawal from chronic use of high doses of marijuana causes physical signs including headache, shakiness, sweating, and stomach pains and nausea.

Withdrawal symptoms also include behavioral signs such as:

- Restlessness, irritability, sleep difficulties, and decreased appetite

What are its overdose effects?

No deaths from overdose of marijuana have been reported.

Which drugs cause similar effects?

Hashish and hashish oil are drugs made from the cannabis plant that are like marijuana, only stronger.

Hashish (hash) consists of the THC-rich resinous material of the cannabis plant, which is collected, dried, and then compressed into a variety of forms, such as balls, cakes, or cookie like sheets. Pieces are then broken off, placed in pipes or mixed with tobacco and placed in pipes or cigarettes, and smoked.



Leaf of marijuana plant

The main sources of hashish are the Middle East, North Africa, Pakistan, and Afghanistan.

Hashish Oil (hash oil, liquid hash, cannabis oil) is produced by extracting the cannabinoids from the plant material with a solvent. The color and odor of the extract will vary, depending on the solvent used. A drop or two of this liquid on a cigarette is equal to a single marijuana joint. Like marijuana, hashish and hashish oil are both Schedule I drugs.

What is its legal status in the United States?

Marijuana is a Schedule I substance under the Controlled Substances Act, meaning that it has a high potential for abuse, no currently accepted medical use in treatment in the United States, and a lack of accepted safety for use under medical supervision.

Although some states within the United States have allowed the use of marijuana for medicinal purpose, it is the U.S. Food and Drug Administration that has the federal authority to approve drugs for medicinal use in the U.S. To date, the FDA has not approved a marketing application for any marijuana product for any clinical indication. Consistent therewith, the FDA and DEA have concluded that marijuana has no federally approved medical use for treatment in the U.S. and thus it remains as a Schedule I controlled substance under federal law.

Marinol, a synthetic version of THC, the active ingredient found in the marijuana plant, can be prescribed for the control of nausea and vomiting caused by chemotherapeutic agents used in the treatment of cancer and to stimulate appetite in AIDS patients. Marinol is a Schedule III substance under the Controlled Substances Act.

Marijuana Concentrates

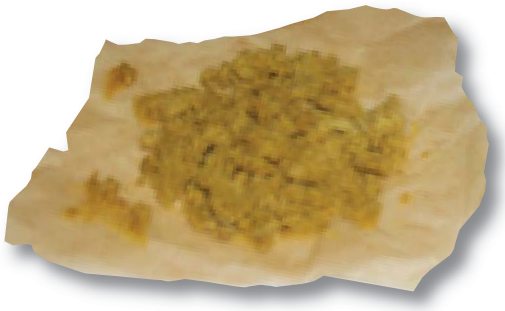
Also Known As: THC Extractions

WHAT ARE MARIJUANA CONCENTRATES?

A marijuana concentrate is a highly potent THC concentrated mass that is most similar in appearance to either honey or butter, which is why it is referred to or known on the street as "honey oil" or "budder."

WHAT IS ITS ORIGIN?

Marijuana concentrates contain extraordinarily high THC levels that could range from 40 to 80 percent. This form of marijuana can be up to four times stronger in THC content than high grade or top shelf marijuana, which normally measures around 20 percent THC levels.



Marijuana concentrate

Many methods are utilized to convert or "manufacture" marijuana into marijuana concentrates. One method is the butane extraction process. This process is particularly dangerous because it uses highly flammable butane to extract the THC from the cannabis plant. Given the extremely volatile nature of butane, this process has resulted in violent explosions. THC extraction labs are being reported nationwide, particularly in the western states and in states where local and state marijuana laws are more relaxed.

What are common street names?

Common street names include:

- 710 (the word "OIL" flipped and spelled backwards), wax, ear wax, honey oil, budder, butane hash oil, butane honey oil (BHO), shatter, dabs (dabbing), black glass, and errl.

What does it look like?

Marijuana concentrates are similar in appearance to honey or butter and are either brown or gold in color.



Marijuana concentrate

How is it abused?

One form of abuse occurs orally by infusing marijuana concentrates in various food or drink products; however, smoking remains the most popular form of ingestion by use of water or oil pipes. A disturbing aspect of this emerging threat is the ingestion of concentrates via electronic cigarettes (also known as e-cigarettes) or vaporizers. Many users of marijuana concentrates prefer the e-cigarette/vaporizer because it's smokeless, odorless, and easy to hide or conceal. The user takes a small amount of marijuana concentrate, referred to as a "dab," then heats the substance using the e-cigarette/vaporizer producing vapors that ensures an instant "high" effect upon the user. Using an e-cigarette/vaporizer to ingest marijuana concentrates is commonly referred to as "dabbing" or "vaping."

What are the Effects of Using Marijuana Concentrates?

Being a highly concentrated form of marijuana, the effects upon the user may be more psychologically and physically intense than plant marijuana use. To date, long term effects of marijuana concentrate use are not yet fully known; but, the effects of plant marijuana use are known. These effects include paranoia, anxiety, panic attacks, and hallucinations. Additionally, the use of plant marijuana increases one's heart rate and blood pressure. Plant marijuana users may also experience withdrawal and addiction problems.

IX. Steroids

WHAT ARE STEROIDS?

Anabolic steroids are synthetically produced variants of the naturally occurring male hormone testosterone that are abused in an attempt to promote muscle growth, enhance athletic or other physical performance, and improve physical appearance.

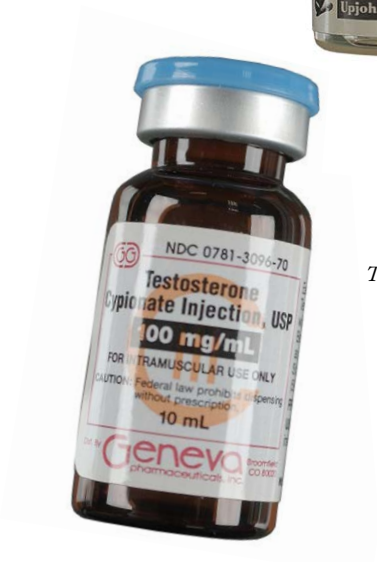
Testosterone, nandrolone, stanozolol, methandienone, and boldenone are some of the most frequently abused anabolic steroids.

WHAT IS THEIR ORIGIN?

Most illicit steroids are smuggled into the U.S. from abroad. Steroids are also illegally diverted from legitimate sources (theft or inappropriate prescribing). The Internet is the most widely used means of buying and selling anabolic steroids. Steroids are also bought and sold at gyms, bodybuilding competitions, and schools from teammates, coaches, and trainers.



Depo-Testosterone



Testosternone Cypionate Injection, USB

What are common street names?

Common street names include:

- Arnolds, Juice, Pumpers, Roids, Stackers, and Weight Gainers

What do they look like?

Steroids are available in:

- Tablets and capsules, sublingual-tablets, liquid drops, gels, creams, transdermal patches, subdermal implant pellets, and water-based and oil-based injectable solutions

The appearance of these products varies depending on the type and manufacturer.

How are they abused?

Steroids are ingested orally, injected intramuscularly, or applied to the skin. The doses abused are often 10 to 100 times higher than the approved therapeutic and medical treatment dosages. Users typically take two or more anabolic steroids at the same time in a cyclic manner, believing that this will improve their effectiveness and minimize the adverse effects.

What is their effect on the mind?

Case studies and scientific research indicate that high doses of anabolic steroids may cause mood and behavioral effects.

In some individuals, steroid use can cause dramatic mood swings, increased feelings of hostility, impaired judgment, and increased levels of aggression (often referred to as "roid rage").

When users stop taking steroids, they may experience depression that may be severe enough to lead one to commit suicide. Anabolic steroid use may also cause psychological dependence and addiction.

What is their effect on the body?

A wide range of adverse effects is associated with the use or abuse of anabolic steroids. These effects depend on several factors including:

- Age, sex, the anabolic steroid used, amount used, and duration of use

In adolescents, anabolic steroid use can stunt the ultimate height that an individual achieves.

In boys, steroid use can cause early sexual development, acne, and stunted growth.

In adolescent girls and women, anabolic steroid use can induce permanent physical changes, such as deepening of the voice, increased facial and body hair growth, menstrual irregularities, male pattern baldness, and lengthening of the clitoris.

In men, anabolic steroid use can cause shrinkage of the testicles, reduced sperm count, enlargement of the male breast tissue, sterility, and an increased risk of prostate cancer.

In both men and women, anabolic steroid use can cause high cholesterol levels, which may increase the risk of coronary artery disease, strokes, and heart attacks. Anabolic steroid use can also cause acne and fluid retention. Oral preparations of anabolic steroids, in particular, can damage the liver.

Users who inject steroids run the risk of contracting various infections due to non-sterile injection techniques, sharing of contaminated needles, and the use of steroid preparations manufactured in non-sterile environments. All these factors put users at risk for contracting viral infections such as HIV/AIDS or hepatitis B or C, and bacterial infections at the sight of injection.

Users may also develop endocarditis, a bacterial infection that causes a potentially fatal inflammation of the heart lining.

What are their overdose effects?

Anabolic steroids are not associated with overdoses. The adverse effects a user would experience develop from the use of steroids over time.

Which drugs cause similar effects?

There are several substances that produce effects similar to those of anabolic steroids. These include human growth hormone (hHG), clenbuterol, gonadotropins, and erythropoietin.

What is their legal status in the United States?

Anabolic steroids are Schedule III substances under the Controlled Substances Act. Only a small number of anabolic steroids are approved for either human or veterinary use. Steroids may be prescribed by a licensed physician for the treatment of testosterone deficiency, delayed puberty, low red blood cell count, breast cancer, and tissue wasting resulting from AIDS.

X. Inhalants

WHAT ARE INHALANTS?

Inhalants are invisible, volatile substances found in common household products that produce chemical vapors that are inhaled to induce psychoactive or mind altering effects.

WHAT IS THEIR ORIGIN?

There are more than 1,000 products that are very dangerous when inhaled — things like typewriter correction fluid, air conditioning refrigerant, felt tip markers, spray paint, air freshener, butane, and even cooking spray. See products abused as inhalants at www.inhalants.org/product.htm (National Inhalant Prevention Coalition).



Highlighter markers



Paint thinner

What are common street names?

Common street names include:

- Gluey, Huff, Rush, and Whippets

What do they look like?

Common household products such as glue, lighter fluid, cleaning fluids, and paint all produce chemical vapors that can be inhaled.

How are they abused?

Although other abused substances can be inhaled, the term "inhalants" is used to describe a variety of substances whose main common characteristic is that they are rarely, if ever, taken by any route other than inhalation.

Inhalants are breathed in through the nose or the mouth in a variety of ways, such as:

- "Sniffing" or "snorting"
- "Bagging" — sniffing or inhaling fumes from substances sprayed or deposited inside a plastic or paper bag
- "Huffing" from an inhalant-soaked rag stuffed in the mouth, or inhaling from balloons filled with nitrous oxide

Inhalants are often among the first drugs that young children use. About 1 in 5 kids report having used inhalants by the eighth grade. Inhalants are also one of the few substances abused more by younger children than by older ones.

What is their effect on the mind?

Inhalant abuse can cause damage to the parts of the brain that control thinking, moving, seeing, and hearing. Cognitive abnormalities can range from mild impairment to severe dementia.

What is their effect on the body?

Inhaled chemicals are rapidly absorbed through the lungs into the bloodstream and quickly distributed to the brain and other organs. Nearly all inhalants produce effects similar to anesthetics, which slow down the body's function. Depending on the degree of abuse, the user can experience slight stimulation, feeling of less inhibition, or loss of consciousness.

Within minutes of inhalation, the user experiences intoxication along with other effects similar to those produced by alcohol. These effects may include slurred speech, an inability to coordinate movements, euphoria, and dizziness. After heavy use of inhalants, users may feel drowsy for several hours and experience a lingering headache.

Additional symptoms exhibited by long-term inhalant users include:

- Weight loss, muscle weakness, disorientation, inattentiveness, lack of coordination, irritability, depression, and damage to the nervous system and other organs

Some of the damaging effects to the body may be at least partially reversible when inhalant abuse is stopped; however, many of the effects from prolonged abuse are irreversible.

Prolonged sniffing of the highly concentrated chemicals in solvents or aerosol sprays can induce irregular and rapid heart rhythms and lead to heart failure and death within minutes. There is a common link between inhalant use and problems in school — failing grades, chronic absences, and general apathy.

Other signs include:

- Paint or stains on body or clothing; spots or sores around the mouth; red or runny eyes or nose; chemical breath odor; drunk, dazed, or dizzy appearance; nausea; loss of appetite; anxiety; excitability; and irritability

What are their overdose effects?

Because intoxication lasts only a few minutes, users try to prolong the high by continuing to inhale repeatedly over the course of several hours, which is a very dangerous practice. With successive inhalations, users may suffer loss of consciousness and/or death.

"Sudden sniffing death" can result from a single session of inhalant use by an otherwise healthy young person. Sudden sniffing death is particularly associated with the abuse of butane, propane, and chemicals in aerosols.

Inhalant abuse can also cause death by asphyxiation from repeated inhalations, which lead to high concentrations of inhaled fumes displacing the available oxygen in the lungs, suffocation by blocking air from entering the lungs when inhaling fumes from a plastic bag placed over the head, and choking from swallowing vomit after inhaling substances.

Which drugs cause similar effects?

Most inhalants produce a rapid high that is similar to the effects of alcohol intoxication.

What is their legal status in the United States?

The common household products that are misused as inhalants are legally available for their intended and legitimate uses. Many state legislatures have attempted to deter youth who buy legal products to get high by placing restriction on the sale of these products to minors.

Even though some substances are not currently controlled by the Controlled Substances Act, they pose risks to individuals who abuse them. The following section describes these drugs of concern and their associated risks.

XI. Drugs of Concern

Even though some substances are not currently controlled by the Controlled Substances Act, they pose risks to individuals who abuse them. The following section describes these drugs of concern and their associated risks.

DXM

WHAT IS DXM?

DXM is a cough suppressor found in more than 120 over-the-counter (OTC) cold medications, either alone or in combination with other drugs such as analgesics (e.g., acetaminophen), antihistamines (e.g., chlorpheniramine), decongestants (e.g., pseudoephedrine), and/or expectorants (e.g., guaifenesin). The typical adult dose for cough is 15 or 30 mg taken three to four times daily. The cough-suppressing effects of DXM persist for 5 to 6 hours after ingestion. When taken as directed, side effects are rarely observed.

WHAT IS ITS ORIGIN?

DXM users can obtain the drug at almost any pharmacy or supermarket, seeking out the products with the highest concentration of the drug from among all the OTC cough and cold remedies that contain it. DXM products and powder can also be purchased on the Internet.



DXM powder

What are common street names?

Common street names include:

- CCC, Dex, DXM, Poor Man's PCP, Robo, Rojo, Skittles, Triple C, and Velvet

What does it look like?

DXM can come in the form of:

- Cough syrup, tablets, capsules, or powder

How is it abused?

DXM is abused in high doses to experience euphoria and visual and auditory hallucinations. Users take various amounts depending on their body weight and the effect they are attempting to achieve. Some users ingest 250 to 1,500 milligrams in a single dosage, far more than the recommended therapeutic dosages described above.

Illicit use of DXM is referred to on the street as "Robo-tripping," "skittling," or "dexing." The first two terms are derived from the products that are most commonly abused, Robitussin and Coricidin HBP. DXM abuse has traditionally involved drinking large volumes of the OTC liquid cough preparations. More recently, however, abuse of tablet and gel capsule preparations has increased.

These newer, high-dose DXM products have particular appeal for users. They are much easier to consume, eliminate the need to drink large volumes of unpleasant-tasting syrup, and are easily portable and concealed, allowing an abuser to continue to abuse DXM throughout the day, whether at school or work.

DXM powder, sold over the Internet, is also a source of DXM for abuse. (The powdered form of DXM poses additional risks to the user due to the uncertainty of composition and dose.)

DXM is also distributed in illicitly manufactured tablets containing

only DXM or mixed with other drugs such as pseudoephedrine and/or methamphetamine.

DXM is abused by individuals of all ages, but its abuse by teenagers and young adults is of particular concern. This abuse is fueled by DXM's OTC availability and extensive "how to" abuse information on various websites.

What is its effect on the mind?

Some of the many psychoactive effects associated with high-dose DXM include:

Confusion, inappropriate laughter, agitation, paranoia, and hallucinations

Other sensory changes, including the feeling of floating and changes in hearing and touch

Long-term abuse of DXM is associated with severe psychological dependence. Abusers of DXM describe the following four dose-dependent "plateaus":

PLATEAU	DOSE (MG)	BEHAVIORAL EFFECTS
1st	100 - 200	Mild stimulation
2nd	200 - 400	Euphoria and hallucinations
3rd	300 - 600	Distorted visual perceptions Loss of motor coordination
4th	500 - 1500	Out-of-body sensations

What is its effect on the body?

DXM intoxication involves:

Over-excitability, lethargy, loss of coordination, slurred speech, sweating, hypertension, and involuntary spasmodic movement of the eyeballs

The use of high doses of DXM in combination with alcohol or other drugs is particularly dangerous, and deaths have been reported. Approximately 5-10 percent of Caucasians are poor DXM metabolizers and at increased risk for overdoses and deaths. DXM taken with antidepressants can be life threatening.

OTC products that contain DXM often contain other ingredients such as acetaminophen, chlorpheniramine, and guaifenesin that have their own effects, such as:

- Liver damage, rapid heart rate, lack of coordination, vomiting, seizures, and coma

To circumvent the many side effects associated with these other ingredients, a simple chemical extraction procedure has been developed and published on the Internet that removes most of these other ingredients in cough syrup.

What are its overdose effects?

DXM overdose can be treated in an emergency room setting and generally does not result in severe medical consequences or death. Most DXM-related deaths are caused by ingesting the drug in combination with other drugs. DXM-related deaths also occur from impairment of the senses, which can lead to accidents.

In 2003, a 14-year-old boy in Colorado who abused DXM died when he was hit by two cars as he attempted to cross a highway. State law enforcement investigators suspect that the drug affected the boy's depth perception and caused him to misjudge the distance and speed of the oncoming vehicles.

Which drugs cause similar effects?

Depending on the dose, DXM can have effects similar to marijuana or ecstasy. In high doses its out-of-body effects are similar to those of ketamine or PCP.

What is its legal status in the United States?

DXM is a legally marketed cough suppressant that is neither a controlled substance nor a regulated chemical under the Controlled Substances Act.

Kratom

WHAT IS KRATOM?

Kratom is a tropical tree native to Southeast Asia. Consumption of its leaves produces both stimulant effects (in low doses) and sedative effects (in high doses), and can lead to psychotic symptoms, and psychological and physiological dependence. The psychoactive ingredient is found in the leaves from the kratom tree. These leaves are subsequently crushed and then smoked, brewed with tea, or placed into gel capsules. Kratom has a long history of use in Southeast Asia, where it is commonly known as thang, kakuam, thom, ketum, and biak. In the U.S., the abuse of kratom has increased markedly in recent years.

How is it abused?

Mostly abused by oral ingestion in the form of a tablet, capsule, or extract. Kratom leaves may also be dried or powdered and ingested as a tea, or the kratom leaf may be chewed.

What are the effects?

At low doses, kratom produces stimulant effects with users reporting increased alertness, physical energy, and talkativeness. At high doses, users experience sedative effects. Kratom consumption can lead to addiction.

Several cases of psychosis resulting from use of kratom have been reported, where individuals addicted to kratom exhibited psychotic symptoms, including hallucinations, delusion, and confusion.

What does it do to your body?

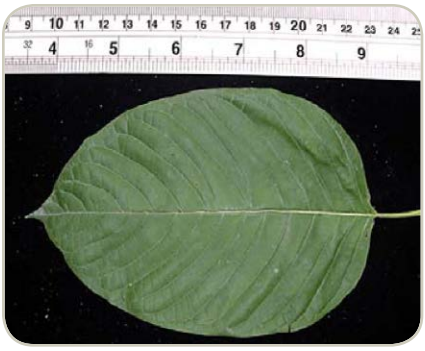
Kratom's effects on the body include nausea, itching, sweating, dry mouth, constipation, increased urination, tachycardia, vomiting, drowsiness, and loss of appetite. Users of kratom have also experienced anorexia, weight loss, insomnia, hepatotoxicity, seizure, and hallucinations.

What is its legal status?

Kratom is not controlled under the Federal Controlled Substances Act; however, there may be some state regulations or prohibitions against the possession and use of kratom. The FDA has not approved Kratom for any medical use. In addition, DEA has listed kratom as a Drug and Chemical of Concern.



Kratom tree



Leaf of kratom tree



Kratom capsules

Salvia Divinorum

WHAT IS SALVIA DIVINORUM?

Salvia divinorum is a perennial herb in the mint family that is abused for its hallucinogenic effects.

WHAT IS ITS ORIGIN?

Salvia is native to certain areas of the Sierra Mazaleca region of Oaxaca, Mexico. It is one of several plants that are used by Mazatec Indians for ritual divination. Salvia divinorum plants can be grown successfully outside of this region. They can be grown indoors and outdoors, especially in humid semitropical climates.

What are common street names?

Common street names include:

- Maria Pastora, Sally-D, and Salvia

What does it look like?

The plant has spade-shaped variegated green leaves that look similar to mint. The plants themselves grow to more than three feet high, have large green leaves, hollow square stems, and white flowers with purple calyces.

How is it abused?

Salvia can be chewed, smoked, or vaporized.

What is its effect on the mind?

Psychic effects include perceptions of bright lights, vivid colors, shapes, and body movement, as well as body or object distortions. Salvia divinorum may also cause fear and panic, uncontrollable laughter, a sense of overlapping realities, and hallucinations.

Salvinorin A is believed to be the ingredient responsible for the psychoactive effects of Salvia divinorum.

What is its effect on the body?

Adverse physical effects may include:

- Loss of coordination, dizziness, and slurred speech

Which drugs cause similar effects?

When Salvia divinorum is chewed or smoked, the hallucinogenic effects elicited are similar to those induced by other Scheduled hallucinogenic substances.

What is its legal status in the United States?

Neither Salvia divinorum nor its active constituent Salvinorin A has an approved medical use in the United States. Salvia is not controlled under the Controlled Substances Act. Salvia divinorum is, however, controlled by a number of states. Since Salvia is not controlled by the CSA, some online botanical companies and drug promotional sites have advertised Salvia as a legal alternative to other plant hallucinogens like mescaline.



Leaves of the salvia divinorum plant

XII. Designer Drugs

Recently, the abuse of clandestinely synthesized drugs has re-emerged as a major worldwide problem. These drugs are illicitly produced with the intent of developing substances that differ slightly from controlled substances in their chemical structure while retaining their pharmacological effects. These substances are commonly known as designer drugs and fall under several drug categories. The following section describes these drugs of concern and their associated risks.

Bath Salts or Designer Cathinones *(Synthetic Stimulants)*

WHAT ARE "BATH SALTS?"

Synthetic stimulants often referred to as "bath salts" are from the synthetic cathinone class of drugs. Synthetic cathinones are central nervous stimulants and are designed to mimic effects similar to those produced by cocaine, methamphetamine, and MDMA (ecstasy). These substances are often marketed as "bath salts," "research chemicals," "plant food," "glass cleaner," and labeled "not for human consumption," in order to circumvent application of the Controlled Substance Analogue Enforcement Act. Marketing in this manner attempts to hide the true reason for the products' existence—the distribution of a psychoactive/stimulant substance for abuse.

WHAT IS THEIR ORIGIN?

Synthetic cathinones are manufactured in East Asia and have been distributed at wholesale levels throughout Europe, North America, Australia, and other parts of the world.



Bath salts

What are common street names?

- Bliss, Blue Silk, Cloud Nine, Drone, Energy-1, Ivory Wave, Lunar Wave, Meow Meow, Ocean Burst, Pure Ivory, Purple Wave, Red Dove, Snow Leopard, Stardust, Vanilla Sky, White Dove, White Knight, White Lightning

What does it look like?

Websites have listed products containing these synthetic stimulants as "plant food" or "bath salts," however, the powdered form is also compressed in gelatin capsules. The synthetic stimulants are sold at smoke shops, head shops, convenience stores, adult book stores, gas stations, and on Internet sites and often labeled "not for human consumption."

How are they abused?

"Bath salts" are usually ingested by sniffing/snorting. They can also be taken orally, smoked, or put into a solution and injected into veins.

What is their effect on the mind?

These synthetic substances are abused for their desired effects, such as euphoria and alertness. Other effects that have been reported from the use of these drugs include psychological effects such as confusion, acute psychosis, agitation, combativeness, aggressive, violent, and self-destructive behavior.

What is their effect on the body?

Adverse or toxic effects associated with the abuse of cathinones, including synthetic cathinones, include rapid heartbeat; hypertension; hyperthermia; prolonged dilation of the pupil of the eye; breakdown of muscle fibers that leads to release of muscle fiber contents into bloodstream; teeth grinding; sweating; headaches; palpitations; seizures; as well as paranoia, hallucinations, and delusions.

What are their overdose effects?

In addition to effects above, reports of death from individuals abusing drugs in this class indicate the seriousness of the risk users are taking when ingesting these products.

Which drugs cause similar effects?

They cause effects similar to those of other stimulants such as methamphetamine, MDMA, and cocaine.

What is their legal status in the United States?

In July 2012, the U.S. Government passed Pub.L. 112- 144, the Synthetic Drug Abuse Prevention Act (SDAPA), that classified a number of synthetic substances under Schedule I of the Controlled Substances Act. SDAPA placed these substances in the most restrictive category of controlled substances. Cannabimimetic agents, including 15 synthetic cannabinoid compounds identified by name, two synthetic cathinone compounds (mephedrone and MDPV), and nine synthetic hallucinogens known as the 2C family, are now restricted by this law. In addition, methylone was permanently controlled by DEA through the administrative process, and another 10 synthetic cathinones became subject to temporary control.

Other synthetic cathinones may be subject to prosecution under the Controlled Substance Analogue Enforcement Act which allows these dangerous substances to be treated as Schedule I controlled substances if certain criteria can be met.

K2 / Spice

WHAT IS K2?

K2 and Spice are just two of the many trade names or brands for synthetic designer drugs that are intended to mimic THC, the main active ingredient of marijuana. These designer synthetic drugs are from the synthetic cannabinoid class of drugs that are often marketed and sold under the guise of "herbal incense" or "potpourri."

Synthetic cannabinoids are not organic, but are chemical compounds created in a laboratory. Since 2009, law enforcement has encountered numerous different synthetic cannabinoids that are being sold as "legal" alternatives to marijuana. These products are being abused for their psychoactive properties and are packaged without information as to their health and safety risks.

Synthetic cannabinoids are sold as "herbal incense" and "potpourri" under names like K2 and Spice, as well as many other names, at small convenience stores, head shops, gas stations, and via the Internet from both domestic and international sources. These products are labeled "not for human consumption" in an attempt to shield the manufacturers, distributors, and retail sellers from criminal prosecution. This type of marketing is nothing more than a means to make dangerous, psychoactive substances widely available to the public.

WHAT IS ITS ORIGIN?

The vast majority of synthetic cannabinoids are manufactured in Asia without manufacturing requirements or quality control standards. The bulk products are smuggled into the United States typically as misbranded imports and have no legitimate medical or industrial use.

What are common street names?

There are numerous and various street names of synthetic cannabinoids as drug manufacturers try to appeal and entice youth and young adults by labeling these products with exotic and extravagant names. Some of the many street names of synthetic marijuana are: "Spice,""K2," Blaze, RedX Dawn, Paradise, Demon, Black Magic, Spike, Mr. Nice Guy, Ninja, Zohai, Dream, Genie, Sence, Smoke, Skunk, Serenity, Yucatan, Fire, and Crazy Clown.



K2/Spice

What does it look like?

These chemical compounds are generally found in bulk powder form, and then dissolved in solvents, such as acetone, before being applied to dry plant material to make the "herbal incense" products. After local distributors apply the drug to the dry plant material, they package it for retail distribution, again without pharmaceutical-grade chemical purity standards, as these have no accepted medical use, and ignoring any control mechanisms to prevent contamination or to ensure a consistent, uniform concentration of the powerful and dangerous drug in each package.

How is it abused?

Spraying or mixing the synthetic cannabinoids on plant material provides a vehicle for the most common route of administration - smoking (using a pipe, a water pipe, or rolling the drug-laced plant material in cigarette papers). In addition to the cannabinoids laced on plant material and sold as potpourri and incense, liquid cannabinoids have been designed to be vaporized through both disposable and reusable electronic cigarettes.

What are its overdose effects?

Overdose deaths have been attributed to the abuse of synthetic cannabinoids, including death by heart attack. Acute kidney injury requiring hospitalization and dialysis in several patients reportedly having smoked synthetic cannabinoids has also been reported by the Centers for Disease Control and Prevention.

Which drugs cause similar effects?

THC, the main psychoactive constituent of marijuana.

What is its effect on the mind?

Acute psychotic episodes, dependence, and withdrawal are associated with use of these synthetic cannabinoids. Some individuals have suffered from intense hallucinations. Other effects include severe agitation, disorganized thoughts, paranoid delusions, and violence after smoking products laced with these substances.

What is its effect on the body?

State public health and poison centers have issued warnings in response to adverse health effects associated with abuse of herbal incense products containing these synthetic cannabinoids. These adverse effects included tachycardia (elevated heart rate), elevated blood pressure, unconsciousness, tremors, seizures, vomiting, hallucinations, agitation, anxiety, pallor, numbness, and tingling. This is in addition to the numerous public health and poison centers which have similarly issued warnings regarding the abuse of these synthetic cannabinoids.

What is its legal status in the United States?

These substances have no accepted medical use in the United States and have been reported to produce adverse health effects. Currently, 26 substances are specifically listed as Schedule I substances under the Controlled Substances Act either through legislation or regulatory action. In addition there are many other synthetic cannabinoids that meet the definition for "cannabimimetic agent" under the Controlled Substances Act and thus are Schedule I substances.

There are many synthetic cannabinoid substances that are being sold as "incense," "potpourri," and other products that are not controlled substances. However, synthetic cannabinoids may be subject to prosecution under the Controlled Substance Analogue Enforcement Act which allows non-controlled drugs to be treated as Schedule I controlled substances if certain criteria can be met. The DEA has successfully investigated and prosecuted individuals trafficking and selling these dangerous substances using the Controlled Substance Analogue Enforcement Act.



K2/Spice

Synthetic Opioids

WHAT ARE SYNTHETIC OPIOIDS?

Synthetic opioids are substances that are synthesized in a laboratory and that act on the same targets in the brain as natural opioids (e.g., morphine and codeine) to produce analgesic (pain relief) effects. In contrast, natural opioids are naturally occurring substances extracted from the seed pod of certain varieties of poppy plants. Some synthetic opioids, such as fentanyl and methadone, have been approved for medical use.

Clandestinely produced synthetic opioids structurally related to the Schedule II opioid analgesic fentanyl were trafficked and abused on the West Coast in the late 1970s and 1980s. In the 1980s, DEA controlled several of these illicitly produced synthetic opioids such as alpha-methylfentanyl, 3-methylthiofentanyl, acetyl-alpha-methylfentanyl, beta-hydroxy-3-methylfentanyl, alpha-methylthiofentanyl, thiofentanyl, beta-hydroxyfentanyl, para-fluorofentanyl, and 3-methylfentanyl.

As of 2013, there has been a re-emergence in the trafficking and abuse of various clandestinely produced synthetic opioids, including several substances related to fentanyl. Some common illicitly produced synthetic opioids that are currently encountered by law enforcement include, but are not limited to, acetyl fentanyl, butyryl fentanyl, beta-hydroxythiofentanyl, furanyl fentanyl, 4-fluoroisobutyryl fentanyl, acryl fentanyl, and U-47700.

WHAT IS THEIR ORIGIN?

Synthetic opioids are believed to be synthesized abroad and then imported into the United States.

What do they look like?

Clandestinely produced synthetic opioids have been encountered in powder form and were identified on bottle caps and spoons, detected within glassine bags, on digital scales, and on sifters which demonstrates the abuse of these substances as replacements for heroin or other opioids. These drugs are also encountered as tablets, mimicking pharmaceutical opioid products. Clandestinely produced synthetic opioids are encountered as a single substance in combination with other opioids (fentanyl, heroin, U-47700) or other substances.



Clandestinely produced counterfeit oxycodone tablets that contain fentanyl.

Opioid powder U-47700.



How are they abused?

Abuse of clandestinely produced synthetic opioids parallels that of heroin and prescription opioid analgesics. Many of these illicitly produced synthetic opioids are more potent than morphine and heroin and thus have the potential to result in a fatal overdose.

What are their effects?

Some effects of clandestinely produced synthetic opioids, similar to other commonly used opioid analgesics (e.g., morphine), may include relaxation, euphoria, pain relief, sedation, confusion, drowsiness, dizziness, nausea, vomiting, urinary retention, pupillary constriction, and respiratory depression.

What are their overdose effects?

Overdose effects of clandestinely produced synthetic opioids are similar to other opioid analgesics. These effects may include stupor, changes in pupillary size, cold and clammy skin, cyanosis, coma, and respiratory failure leading to death. The presence of triad of symptoms such as coma, pinpoint pupils, and respiratory depression are strongly suggestive of opioid poisoning.

Which drugs cause similar effects?

Some drugs that cause similar effects include other opioids such as morphine, hydrocodone, oxycodone, hydromorphone, methadone, and heroin.

What is their legal status in the United States?

Several synthetic opioids are currently controlled under the Controlled Substances Act. Recently, the DEA temporarily placed U-47700 and several other substances that are structurally related to fentanyl, such as acetyl fentanyl, butyryl fentanyl, beta-hydroxythiofentanyl, and furanyl fentanyl, in Schedule I of the Controlled Substances Act. Other synthetic opioid substances may be subject to prosecution under the Controlled Substance Analogue Enforcement Act which allows non-controlled substances to be treated as Schedule I substances if certain criteria are met. The DEA has successfully investigated and prosecuted individuals trafficking and selling these dangerous substances using the Controlled Substances Analogue Enforcement Act.

XIII. Resources

DRUG USE PREVENTION RESOURCES

Drug prevention programs are designed and implemented on many levels. The federal government has instituted a number of national drug prevention programs which reach targeted populations through public service announcements, grant programs, educational programs, and the sharing of expertise. State and local governments also have a significant number of prevention programs that are tailored to address particular problems and needs. Law enforcement and the military have brought drug prevention expertise into classrooms and communities; businesses have also contributed significantly to drug prevention through sponsored programs, drug-free policies, and corporate support for community initiatives. Other segments of society, including faith-based institutions, civic organizations, and private foundations are also active forces in drug prevention.

Below is a partial list of drug prevention agencies and programs. There are many other outstanding efforts which are ongoing across the nation; it is impossible to include them all. Some programs are aimed at particular populations or specific drugs. Within a given agency, there may be many prevention programs which are aimed at different audiences.

FEDERAL DRUG PREVENTION AGENCIES AND PROGRAMS:

Drug Enforcement Administration (DEA):

In addition to dismantling major drug trafficking organizations, DEA is committed to reducing the demand for drugs in America. DEA's Demand Reduction Program is carried out by Special Agents across the United States who work in communities to share expertise and information on drug trends, emerging problems, and the dangers of drugs.

www.dea.gov
www.JustThinkTwice.com
www.GetSmartAboutDrugs.com
www.operationprevention.com

Office of National Drug Control Policy (ONDCP):

This office reports to the President of the United States.

www.whitehousedrugpolicy.gov

Substance Abuse and Mental Health Services Administration (SAMHSA):

This organization is responsible for overseeing and administering mental health, drug prevention, and drug treatment programs around the nation. The Center for Substance Abuse Prevention (CSAP) and the Center for Substance Abuse Treatment (CSAT) are part of SAMHSA.

www.samhsa.gov
www.samhsa.gov/prevention
www.samhsa.gov/about-us/who-we-are/

U.S. Department of Education (ED):

ED's anti-drug program is housed in the Office of Safe and Healthy Students.

www.ed.gov

National Institute on Drug Abuse (NIDA):

NIDA conducts and disseminates the results of research about the effects of drugs on the body and the brain. NIDA is an excellent source of information on drug addiction.

www.nida.nih.gov

Other Anti-Drug Organizations:

National Association of State Alcohol and Drug Abuse Directors (NASADAD)
www.nasadad.org

Community Anti-Drug Coalitions Of America (CADCA)
www.cadca.org

National Crime Prevention Council (NCPC)
www.ncpc.org

National Families in Action (NFIA)
www.nationalfamilies.org

You can obtain free anti-drug information from:

Substance Abuse and Mental Health Services Administration
www.store.samhsa.gov

The National Center on Addiction and Substance Abuse at Columbia University (CASA)
www.casacolumbia.org

Elks Drug Awareness Program
www.elks.org/dap

Partnership for Drug-Free Kids
www.drugfree.org

American Council for Drug Education (ACDE)
www.acde.org

Drug Strategies
www.drugstrategies.org

Youth Anti-Drug Organizations:

Young Marines
www.youngmarines.com

Drug Abuse Resistance Education (DARE)
www.dare.com

Students Against Destructive Decisions (SADD)
www.sadd.org

Law Enforcement Exploring
exploring.learningforlife.org/services/career-exploring/law-enforcement/


GET THE FACTS ABOUT DRUGS
JUST THINK TWICE
A Resource for Teens
www.justthinktwice.com


GET SMART
ABOUT DRUGS
A DEA RESOURCE FOR PARENTS, EDUCATORS & CAREGIVERS
WWW.GETSMARTABOUTDRUGS.COM


Follow DEA
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@DEAHQ


OPERATION
PREVENTION
The Science Behind
Opioid Addiction
www.operationprevention.com

Module-4
Drugs of Abuse

CLASSIFICATION
CANNABIS
COCAINE
OPIOIDS
DEPRESSANTS

Classification of Drugs of Abuse

Based on origin

- Naturally occurring
- Semi-synthetic
- Synthetic

Based on acidic/basic nature

- Acidic drugs
- Basic drugs

Based on Usages

- Predator drugs
- Club drugs
- Human performance altering drugs
- Inhalants

Based on chemistry of Active compound

- Alkaloid
- Cannabinoids
- Glycosides
- Toxalbumins
- Others

Classification

- Based on Mode of Action

1. Narcotics/Analgesics
2. Stimulants
3. Depressants
4. Hallucinogens
5. Anabolic Steroids

Classification by Schedules/Regulations

Schedule	Medical Use	Controls on Prescriptions	Required Security	Potential for Abuse	Addiction Potential	Examples
I	None accepted	None; used only for research	Vault or safe	Highest	Severe	LSD, heroin, MDMA (Ecstasy), GHB
II	Some accepted uses with restrictions	Written prescription with no refills	Vault or safe		Severe	Morphine and many related opiates, cocaine, amphetamine and methamphetamine, most barbiturates, oxycodone
III	Accepted uses	Written or oral (phone in), limits on refills and time	Secured area		Moderate to low	Ketamine, anabolic steroids, some codeine preparations
IV	Accepted uses	Written or oral (phone in), limits on refills and time	Secured area		Limited	Benzodiazepines such as Valium, mild sleep aids
V	Accepted uses	Over the counter, written, or oral (phone in), limits on refills and time	Secured area	Lowest	Limited	Selected preparations of codeine

What Are Neurotransmitters?

- Neurotransmitters are **chemical messengers** that transmit signals across a **synapse** (the junction between neurons or between a neuron and another target cell — such as a muscle or gland).

- They are released from **presynaptic neurons** in response to an electrical impulse (action potential).

- They bind to **specific receptors** on the **postsynaptic cell**, causing excitation or inhibition of that cell.

- **Key Characteristics**
  - Synthesized and stored in neurons.
  - Released in response to nerve stimulation.
  - Produce specific effects on postsynaptic cells.
  - Have mechanisms for removal or degradation (reuptake or enzymatic breakdown)

Neurotransmitter	Major Functions	Clinical / Behavioral Correlation
Acetylcholine (ACh)	Muscle activation, learning, memory	↓ ACh → Alzheimer's disease, paralysis; ↑ → muscle spasms
Dopamine (DA)	Motivation, reward, movement, mood	↓ DA → Parkinson's disease; ↑ → Schizophrenia, addiction
Norepinephrine (NE)	Arousal, attention, "fight or flight" response	↓ NE → Depression; ↑ → Anxiety, hypertension
Epinephrine (Adrenaline)	Stress response, cardiac stimulation	Excess → anxiety, insomnia
Serotonin (5-HT)	Mood regulation, sleep, appetite	↓ → Depression, aggression, insomnia; ↑ → Serotonin syndrome
Gamma-aminobutyric acid (GABA)	Main inhibitory neurotransmitter	↓ → Anxiety, seizures, epilepsy
Glutamate	Major excitatory neurotransmitter, learning and memory	↑ → Excitotoxicity, stroke damage

NEUROTRANSMITTERS

ADRENALINE
fight or flight
produced in stressful situations. Increases heart rate and blood flow, leading to physical boost and heightened awareness.

GABA
calming
Calms firing nerves in the central nervous system. High levels improve focus, low levels cause anxiety. Also contributes to motor control and vision.

NORADRENALINE
concentration
affects attention and responding actions in the brain. Contracts blood vessels, increasing blood flow.

ACETYLCHOLINE
learning
Involved in thought, learning and memory. Activates muscle action in the body. Also associated with attention and awakening.

DOPAMINE
pleasure
feelings of pleasure, also addiction, movement and motivation. People repeat behaviors that lead to dopamine release.

GLUTAMATE
memory
Most common neurotransmitter. Involved in learning and memory, regulates development and creation of nerve contacts.

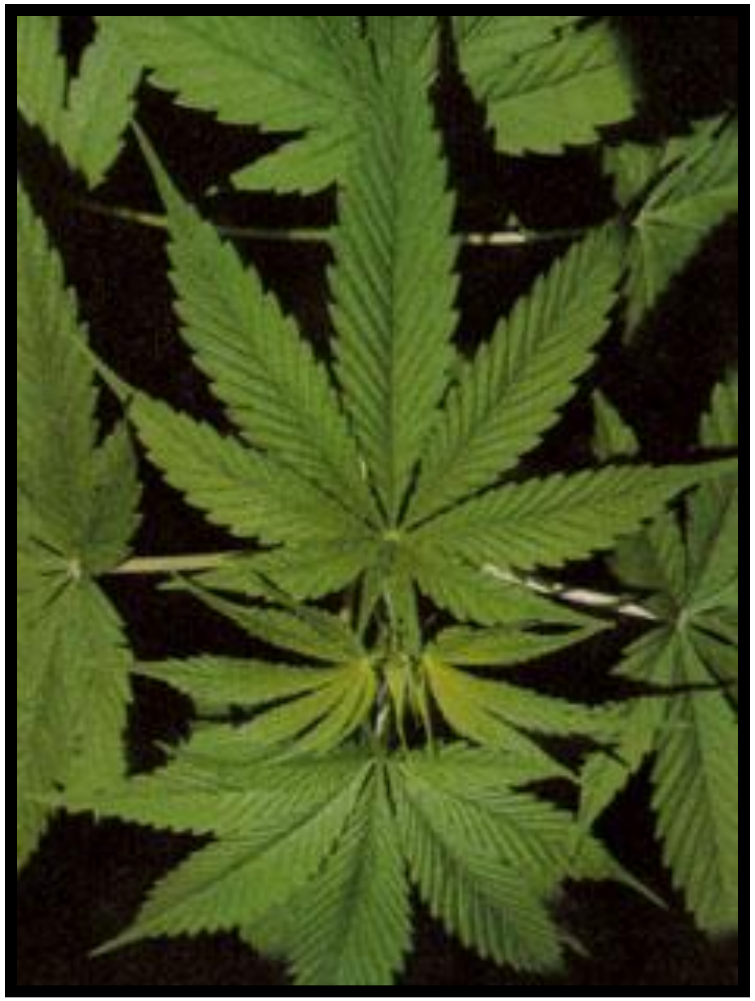
SEROTONIN
mood
contributes to well-being and happiness. Helps sleep cycle and digestive system regulation. Affected by exercise and light exposure.

ENDORPHINS
euphoria
Released during exercise, excitement and sex, producing well-being and euphoria, reducing pain

Drugs of Abuse

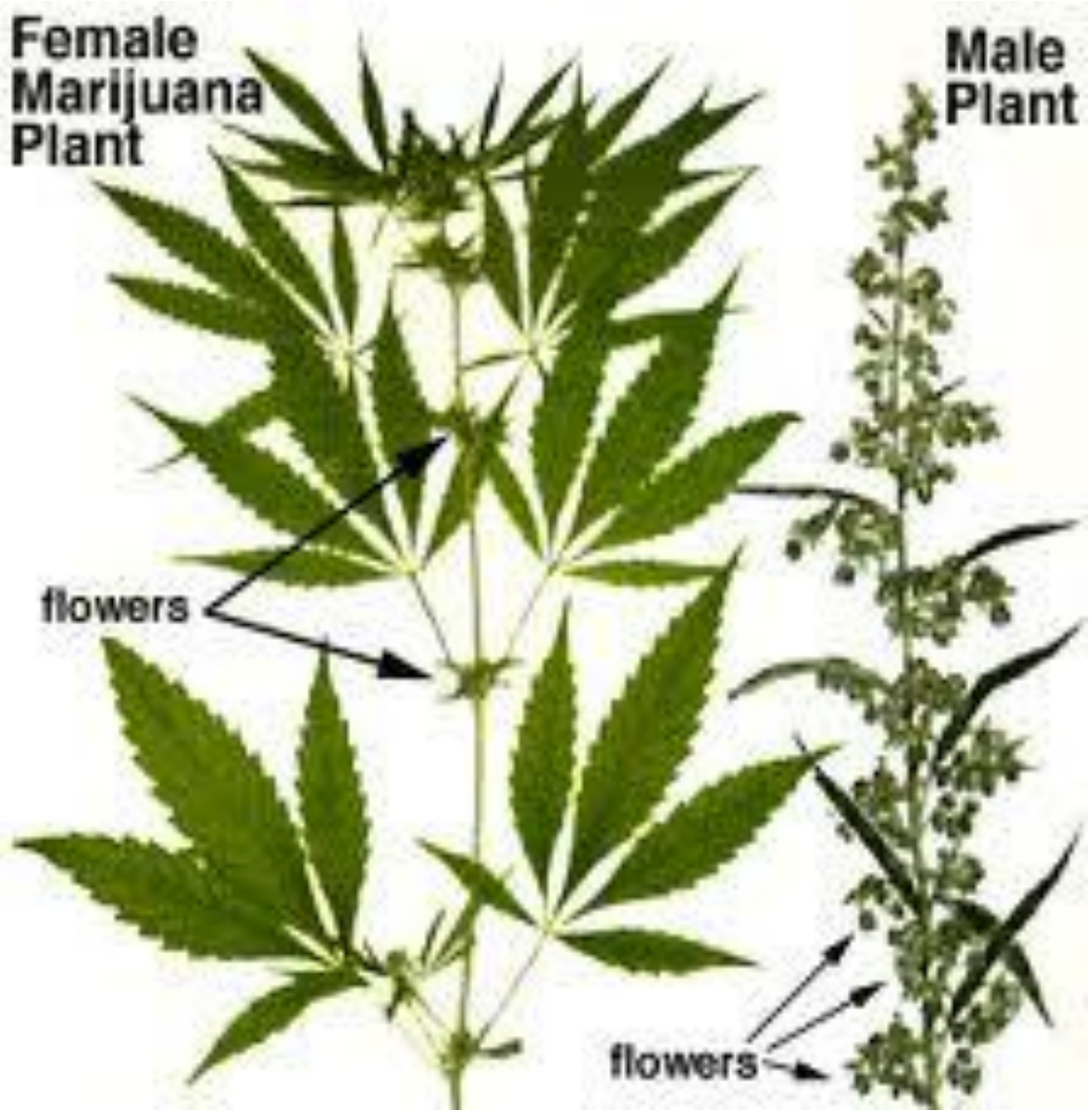


Common names include - hemp, marihuana, marijuana, pot, gandia, grass, chanvre etc.





•Cannabis is an annual, dioecious, flowering herb.



Introduction

- **Scientific Name:** *Cannabis sativa*, *Cannabis indica*, and *Cannabis ruderalis*
- **Common Names:** Marijuana, Ganja, Bhang, Charas, Hashish, Weed, Pot, Dope.
- **Type of Drug: Psychoactive drug** (depressant with mild hallucinogenic effects).
- **Source:** Dried leaves, flowering tops, seeds, and resin of the *Cannabis* plant.
- **Relevance in Forensic Science**
  - Cannabis is one of the most **commonly abused drugs worldwide**.
  - Identified under **Schedule I** substances in many countries.
  - Important for forensic chemists in **drug identification, quantification, and toxicological analysis**.

Origin

- The term, **marijuana**, is believed to be originated from a Portuguese or Spanish word for intoxicating.
- It is present in the plant, Cannabis sativa, which is also known as hemp.
- Hemp is commercially valuable for more than marijuana; its stem yields a fiber which can be made into ropes.
- All plants of the genus Cannabis produce a complex chemical called **delta9-tetrahydrocannabinol (THC).**

Active Constituents

Major Psychoactive Compound

A. Δ^9-Tetrahydrocannabinol (Δ^9-THC):

- Principal psychoactive compound.
- Lipid-soluble; easily crosses the blood–brain barrier.
- Concentration varies depending on the part of plant and preparation.

B. Other Cannabinoids

Compound	Activity	Notes
Cannabidiol (CBD)	Non-psychoactive	Modulates THC effects; therapeutic potential (anti-anxiety, anti-inflammatory).
Cannabinol (CBN)	Mildly psychoactive	Oxidation product of THC; found in aged cannabis
Cannabichromene (CBC)	Non-psychoactive	Contributes to entourage effect.
Cannabigerol (CBG)	Non-psychoactive	Biosynthetic precursor of other cannabinoids.

C. Other Constituents

- Terpenes, flavonoids, alkaloids, and plant waxes contribute to aroma and pharmacological diversity.

Active Constituents/ Principles

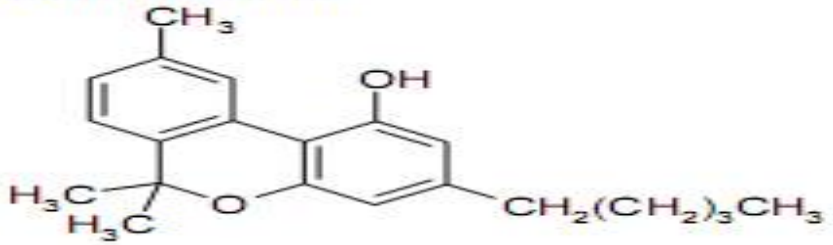
1. Cannabinol
2. Cannbidiol
3. Cannabidiolic acid
4. Tetrahydrocannabinol (THC)

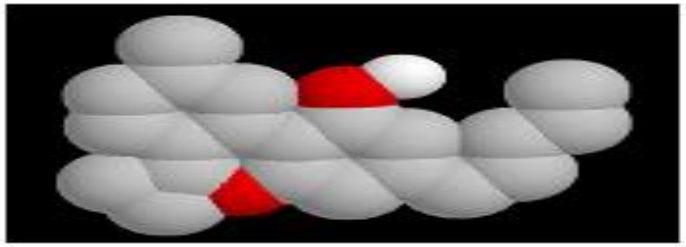
- THC exists in isomeric forms; Δ^9 THC is the most potent psychoactive component.
- Cannabinoids:
  - They are **not considered alkaloids** because they do not contain nitrogen group.
  - Absorbed by fatty tissues in organs of the body (including brain)
  - Can be detected in the body for days, but can be detected weeks later in chronic users.

- Active constituents are found in
  - Leaves & fruiting tops of female plants
  - Dried leaves & fruiting shoots of both male & female plants
  - Cannabis seeds are a less well known though potent source of Ω-3-fatty acids

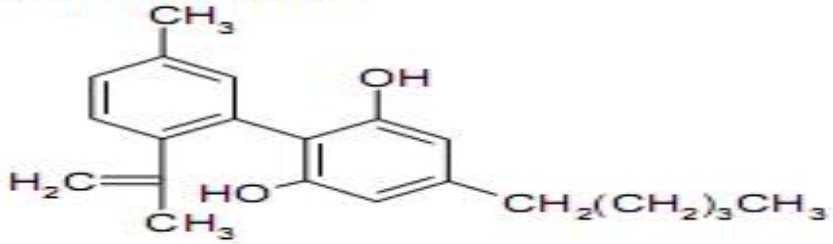
Cannabinoids

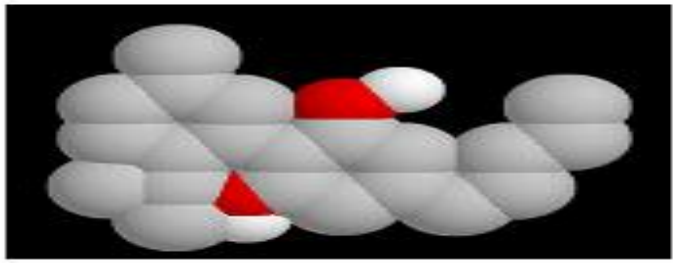
Cannabinol



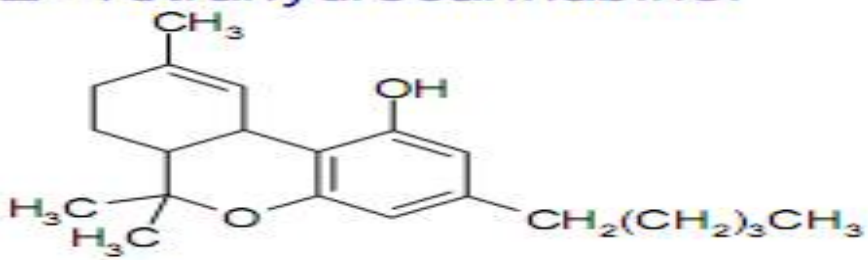


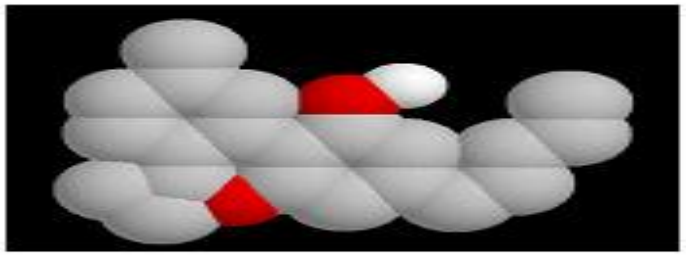
Cannabidiol





Δ^9-Tetrahydrocannabinol





$C_{21}H_{30}O_2$ Tetrahydro 6,6,9-trimethyl-3-pentyl-6H-dibenzo [b,d] pyran-1-ol

Preparations

- Concentration of THC varies in different parts of cannabis plant-
  - Resinous extract > flowers > leaves > stem/roots > seeds(almost nil).
- Based on the relative proportion of THC, different preparations are as-
  a. Bhang
  b. Ganja
  c. Charas
  d. Hashish oil

Cannabis Preparations

Preparation	Source/Description	THC Content	Common Region/Use
Bhang	Dried leaves and small stems (lowest potency)	2–5%	India — consumed as drink or sweet during festivals.
Ganja (Marijuana)	Dried flowering tops and leaves of female plants	5–10%	Commonly smoked.
Charas (Hashish)	Resinous exudate from flower tops	10–20%	High potency; smoked or mixed with tobacco.
Hashish Oil (Cannabis Oil)	Extracted resin dissolved in solvent	30–60%	Most potent form; used for smoking or ingestion.
Sinsemilla	Flowering tops of unfertilized female plants	15–25%	Cultivated for higher THC yield.

Bhang

- **Preparation**: Bhang is a traditional preparation of cannabis in India, where the leaves and flowering shoots of both male & female cannabis plant are ground into a paste with water or milk.

- **Consumption**: It is often consumed as a beverage, either mixed with milk (bhang lassi) or in a sweetened drink (bhang thandai). **Not used for smoking.**

- **Effects**: Bhang consumption is usually milder compared to other forms, providing a more subtle and relaxing high. It has been traditionally used in cultural and religious contexts in India.

- Ganja Marijuana:
  - **Preparation**: Ganja refers to the dried top leaves and unfertilized flower of young female cannabis plant.
  - **Consumption**: It is commonly smoked in hand-rolled cigarettes (joints) or pipes, but it can also be vaporized or infused into edibles.
  - **Effects**: Ganja produces a range of effects depending on the strain and THC content. Users typically experience euphoria, relaxation, altered perception, increased appetite, and impaired coordination. It is one of the most commonly abused forms of cannabis worldwide.

- Charas:
  - **Preparation**: Charas is a potent form of cannabis resin, made by hand-rubbing the flowering tops of the cannabis plant.
  - **Appearance**: It is dark brown to black in color and has a sticky and pliable texture.
  - **Consumption**: Charas is typically smoked in pipes, mixed with tobacco in cigarettes (joints), or vaporized.
  - **Effects**: Due to its high THC content, charas produces strong psychoactive effects. Users may experience intense euphoria, relaxation, and altered perception. It is commonly abused for its potent effects.

- **Hash** (Hashish):
  - **Preparation**: Hash is made by collecting and compressing the resin glands (trichomes) from the cannabis plant.
  - **Appearance**: It can range in color from light brown to black, with a texture that can be solid or pliable.
  - **Consumption**: Hash is typically smoked in pipes, mixed with tobacco in cigarettes (joints), or vaporized.
  - **Effects**: Hash is more potent than traditional marijuana (ganja) due to its higher concentration of THC. Users may experience strong euphoria, relaxation, and altered perception. It is widely abused for its concentrated psychoactive effects.

- **Hash Oil** (Also known as Cannabis Oil or Honey Oil):
  - **Preparation**: Hash oil is a highly concentrated extract of cannabis, made using solvents such as butane or CO2.
  - **Appearance**: It is usually thick and viscous, with a dark color.
  - **Consumption**: Hash oil can be used in vaporizers, mixed with dried herbs for smoking, or consumed orally.
  - **Effects**: Hash oil is extremely potent due to its high THC content. Users may experience intense euphoria, relaxation, and sedation. It is commonly used for both recreational and medical purposes.

Mode of Intake

Mode of Intake	Example / Form	Absorption & Onset	Duration of Effect	Remarks
Smoking / Inhalation	Ganja, Charas, Marijuana (rolled joints, pipes, chillums)	Rapid absorption via lungs; onset within minutes	1–3 hours	Most common recreational route; bioavailability ~25–30%.
Oral Ingestion	Bhang, edibles, cannabis-infused oils	Slow absorption via GI tract; onset 30–90 min	4–8 hours	Effects last longer but less predictable.
Vaporization	Electronic vape devices	Rapid, clean delivery	1–3 hours	Increasingly popular; fewer combustion by-products.
Sublingual / Buccal	Tinctures, sprays	Moderate absorption through oral mucosa	1–3 hours	Used in medical marijuana formulations.

Signs & Symptoms

- Acts as a hallucinogen, initially causing excitement.
- Low or social dose causes sense of well being with initial restlessness and hilarity followed by dreamy, carefree sense of relaxation, hallucinations, disorientation of time & space, sense of hunger (particularly for sweets).

- Overall effects depends upon-
  1. Personality of the subject
  2. Whether subject is accustomed to consumption of cannabis
  3. Environment at the time of consumption

- <u>In a peaceful environment-</u>
  - Visual hallucination of colour & form
  - Visual fantasy, often with sexual content
  - Flushing, red eye, dilated pupils, mental confusion & lassitude
  - Disorientation of time, place & person, tremors, jerky movements
  - Uneasy sleep, hangover may persist for a day

Short-term Effects of Cannabis Abuse

1. **Euphoria and Relaxation**: Cannabis use often results in feelings of euphoria, relaxation, and a sense of well-being.
2. **Altered Perception**: Users may experience changes in perception, including altered time perception, heightened sensory perception, and distortions in visual and auditory perceptions.
3. **Impaired Coordination and Motor Skills**: Cannabis can impair coordination and motor skills, leading to decreased balance, slowed reaction times, and difficulty performing tasks requiring precision.
4. **Increased Heart Rate and Blood Pressure**: Cannabis use can cause a temporary increase in heart rate and blood pressure, which may lead to palpitations and feelings of anxiety or paranoia.
5. **Dry Mouth and Red Eyes**: Commonly known as "cottonmouth," cannabis use can result in dry mouth due to decreased saliva production. It can also cause redness and irritation of the eyes.
6. **Increased Appetite**: Cannabis use often leads to an increase in appetite, commonly referred to as "the munchies."
7. **Anxiety and Paranoia**: Some individuals may experience feelings of anxiety, paranoia, or panic attacks, particularly at higher doses or in susceptible individuals.
8. **Memory and Cognitive Impairment**: Cannabis can impair short-term memory and cognitive function, making it difficult to concentrate, learn new information, or perform complex tasks.

Long-term Effects of Chronic Cannabis Abuse

1. **Cognitive Impairment**: Long-term cannabis abuse has been associated with persistent cognitive impairment, including deficits in memory, attention, and executive function. Chronic use during adolescence may have a more pronounced impact on cognitive development.
2. **Psychiatric Disorders**: Chronic cannabis abuse has been linked to an increased risk of developing psychiatric disorders, including schizophrenia, depression, and anxiety disorders. However, the causal relationship is complex and not fully understood.
3. **Respiratory Issues**: Smoking cannabis can lead to respiratory problems, such as chronic bronchitis, coughing, and increased mucus production. Long-term use may also increase the risk of respiratory infections and lung cancer.
4. **Dependence and Addiction**: Long-term cannabis abuse can lead to the development of dependence and addiction, characterized by cravings, tolerance, withdrawal symptoms (such as irritability, anxiety, and insomnia), and difficulty controlling use despite negative consequences.
5. **Social and Behavioral Problems**: Chronic cannabis abuse can contribute to social and behavioral problems, including impaired relationships, decreased motivation, educational or occupational difficulties, and legal problems related to drug use.
6. **Impact on Brain Structure and Function**: Long-term cannabis abuse has been associated with alterations in brain structure and function, particularly in regions involved in memory, emotion regulation, and decision-making. These changes may persist even after cessation of use.
7. **Negative Impact on Mental Health**: Chronic cannabis abuse has been linked to an increased risk of developing mood disorders, psychotic disorders, and other mental health problems, particularly in vulnerable individuals.

Long-term Effects of Chronic Cannabis Abuse

1. **Cognitive Impairment**:
    1. **Memory**: Chronic cannabis use can impair short-term memory and cognitive function.
    2. **Learning**: Difficulty in acquiring new information and learning tasks.
    3. **Attention**: Reduced ability to focus and sustain attention.
2. **Respiratory Issues**:
    1. **Chronic Bronchitis**: Smoking cannabis can lead to chronic bronchitis-like symptoms, including coughing and wheezing.
    2. **Increased Risk of Lung Infections**: Due to the effects of smoke on the respiratory system.
3. **Mental Health Effects**:
    1. **Depression**: Chronic cannabis use has been linked to an increased risk of depression.
    2. **Anxiety Disorders**: Prolonged use can exacerbate anxiety disorders and lead to panic attacks.
    3. **Psychosis**: Individuals with a predisposition to psychotic disorders may experience worsened symptoms.
4. **Addiction and Withdrawal**:
    1. **Cannabis Use Disorder**: Chronic abuse can lead to dependence, characterized by cravings and difficulty quitting despite negative consequences.
    2. **Withdrawal Symptoms**: Upon cessation of use, some individuals may experience withdrawal symptoms such as irritability, insomnia, loss of appetite, and mood swings.
5. **Social and Occupational Impairment**:
    1. **Relationships**: Chronic use can strain relationships due to changes in behavior and priorities.
    2. **Work/School**: Impaired performance in work or school due to cognitive deficits and motivation issues.
6. **Cardiovascular Effects**:
    1. **Increased Heart Rate**: Acute use can lead to a temporary increase in heart rate, potentially risky for individuals with heart conditions.
    2. **Cardiovascular Disease Risk**: Chronic use may increase the risk of cardiovascular diseases in susceptible individuals.
7. **Endocrine Effects**:
    1. **Hormonal Imbalance**: Long-term cannabis use may disrupt hormone production, affecting reproductive health and menstrual cycles.
    2. **Male Fertility**: Some studies suggest a link between chronic cannabis use and reduced sperm count and quality.
8. **Immune System**:
    1. **Suppression**: Chronic use may weaken the immune system, making individuals more susceptible to infections.

Forensic Analysis of Cannabis

Physical Examination

- The methods used to identify cannabis products depend upon the nature of the product.
- Herbal material can be identified based on its morphological characteristics alone, provided that the required ones are present.
  - The cannabis sample is visually examined for its color, texture, and form (e.g., leaves, buds, resin).
  - Any packaging or containers are inspected for labels, markings, or other identifying features.
- Where there are no morphological characteristics, as in the case of resin and hashish oil, the identification is based on chemical analysis, demonstrating the presence of cannabinoids, such as tetrahydrocannabinol (THC), its degradation product cannabinol (CBN) and/or cannabidiol (CBD).



Dorsal aspects of leaves



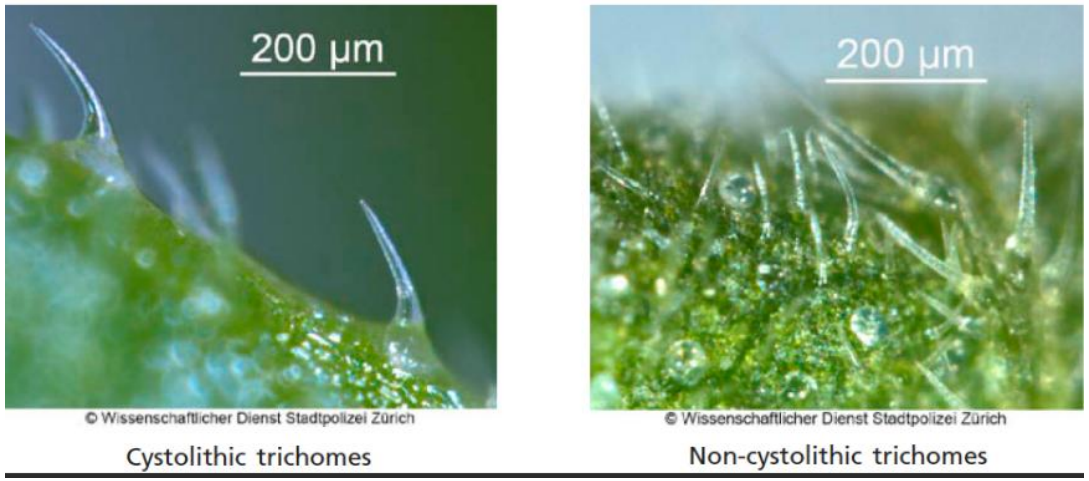
Magnified view of trichomes on leaves of cannabis



Fruiting buds of male plant

Microscopic Examination

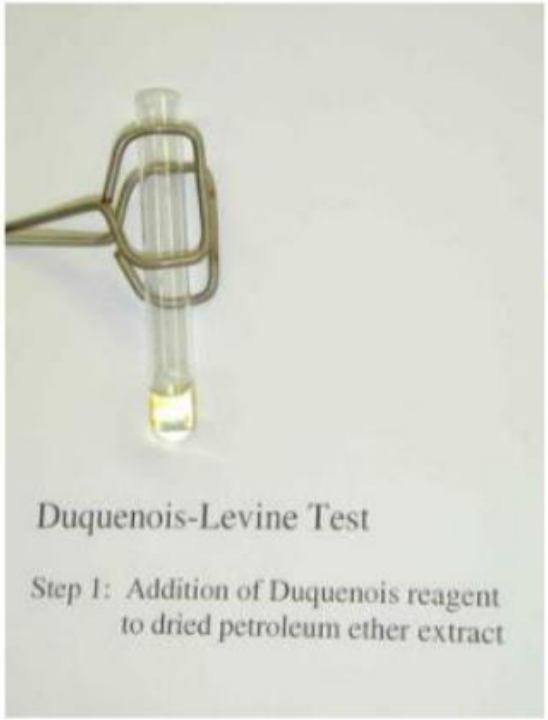
- Crude form of *Cannabis sativa* is examined under the microscope-
  - Non glandular hair **(trichomes)**, numerous, unicellular, rigid, curved, with a slender pointed apex and an enlarged base, usually containing a **cystolithic hair** but frequently broken and the cystolith freed (especially in cannabis resin)
  - Presence of fragments of bracts, bractioles, and other parts of plant

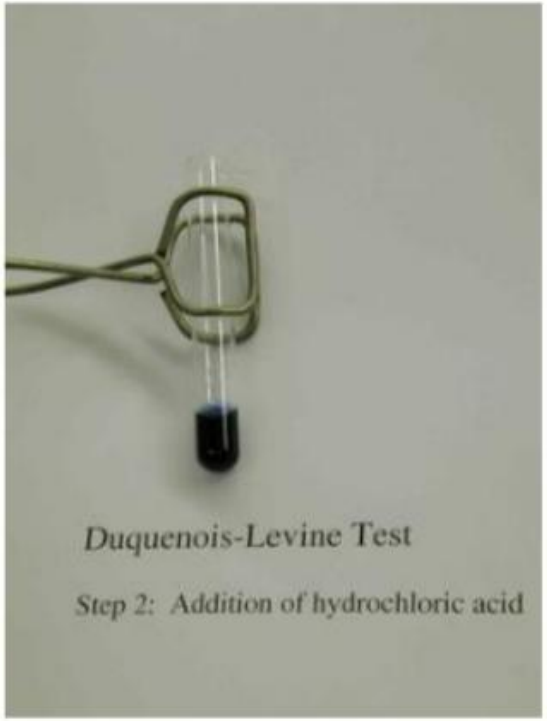


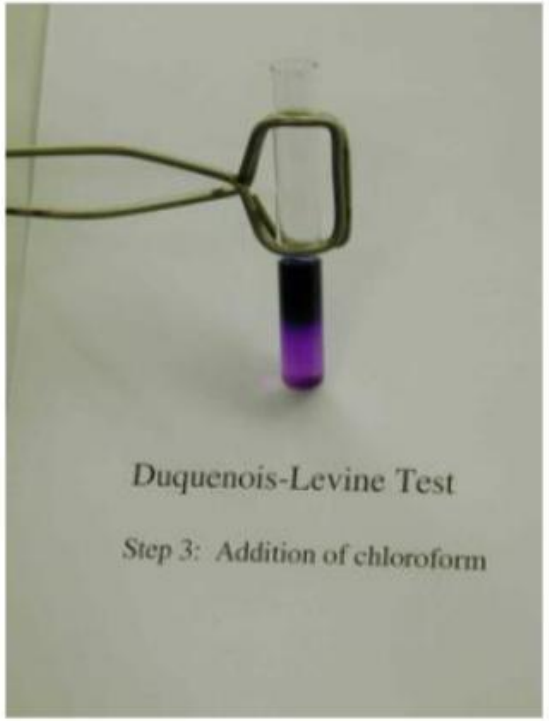
Colour Tests for *Cannabis sativa*

1. **Fast Blue B Test:**
   - Solid Fast Blue B reagent; solid Fast Blue B(2.5) : anhydrous sodium sulphate (100), is added to small amount of residue of extract in a test tube.
   - 1 ml of chloroform is added & shaken and kept for 2 minutes.
   - The chloroform layer becomes purple red in colour.

2. **Duquenois-Levine Test:**
   - 2 ml of reagent (5 drops of Acetaldehyde & 0.4 gm of Vanillin dissolved in 20 ml of 95% ethanol) is added to residue of extract in a test tube and then shaken for 1 minutes.
   - 2 ml of conc HCl is added, shaken and allowed to stand for 10 minutes
   - If colour develops, 2 ml of chloroform is added
   - The lower (chloroform) layer becomes violet.


Duquenois-Levine Test
Step 1: Addition of Duquenois reagent
to dried petroleum ether extract


Duquenois-Levine Test
Step 2: Addition of hydrochloric acid


Duquenois-Levine Test
Step 3: Addition of chloroform

Colour Tests...

3. Beam's Alkaline Test:
   - 3-4 drops of ethanolic KOH are added to extracted residue in a porcelain dish.
   - A positive reaction is indicated by presence of violet colour after few minutes

4. Negm's Test
5. Ghamrawy's Test
6. Gibb's Test
7. Bouquet's Test

Test for differentiation between Bhang, Ganja and Charas

- **Reagent 1**: p-Aminophenol (1 mg) in ethanol (10 ml)
- **Reagent 2**: Caustic potash (1 g) in distilled water (10 ml)

- Extract the suspected material of cannabis in ethanol.

- Take a drop of extract in a cavity of a spot tile or in a micro tube, add 2 drops of **reagent 1** and mix thoroughly followed by addition of 2 drops of **reagent 2**.
  - Bhang gives green colour,
  - Ganja gives blue colour,
  - Charas gives violet colour.

TLC

- Solvent system:

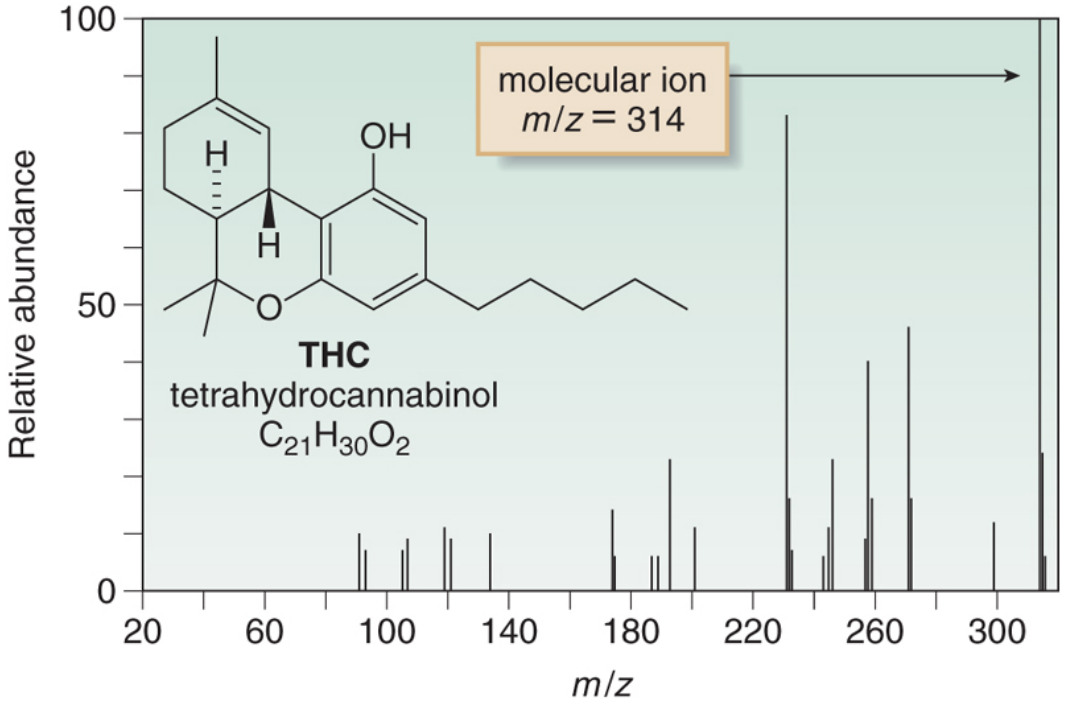
1. Benzene(25) : n-Hexane(10) : Diethylamine(1)
2. n-Hexane(84) : Dichloro-methane(6) : Acetone(60)

- HRf values
  - Cannabidiol – 45, 60
  - Cannabinol – 25, 54
  - THC – 35, 63

- Spraying Reagent:
  - Fast Blue B (by dissolving 15 mg of salt in 20 ml of 0.1N NaOH solution)
  - Violet/ bluish colour of spots

GC Data for Cannabinoids

- Column & packing material:
  - 6’ X 4mm (ID) glass column, packed with 2-3/100 SE-30 on chromosorb W, 80/100 mesh
  - Sample volume: 1-3µl
- Carrier gas-
  - Nitrogen; 60 ml/min
- Column temperature: 240^0C
- Detector & injection temperature: 270^0C
- Detector: FID
- Retention time: (in minutes)
  - Cannabinol – 11.0
  - Cannabidiol – 6.0
  - THC – 8.0



Cocaine

Classification by Schedules/Regulations

Schedule	Medical Use	Controls on Prescriptions	Required Security	Potential for Abuse	Addiction Potential	Examples
I	None accepted	None; used only for research	Vault or safe	Highest	Severe	LSD, heroin, MDMA (Ecstasy), GHB
II	Some accepted uses with restrictions	Written prescription with no refills	Vault or safe		Severe	Morphine and many related opiates, cocaine, amphetamine and methamphetamine, most barbiturates, oxycodone
III	Accepted uses	Written or oral (phone in), limits on refills and time	Secured area		Moderate to low	Ketamine, anabolic steroids, some codeine preparations
IV	Accepted uses	Written or oral (phone in), limits on refills and time	Secured area		Limited	Benzodiazepines such as Valium, mild sleep aids
V	Accepted uses	Over the counter, written, or oral (phone in), limits on refills and time	Secured area	Lowest	Limited	Selected preparations of codeine





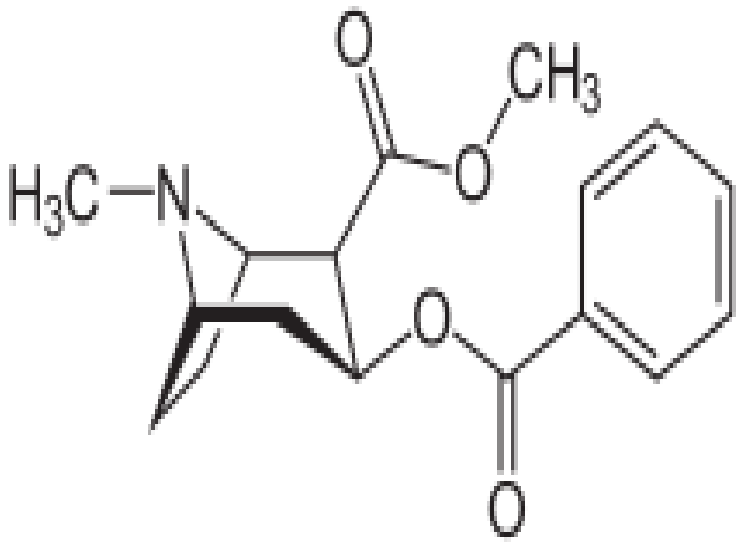


Coca Alkaloids

- Alkaloids found in coca plant, *Erythroxyum coca* are either **tropane** type or **pyrrolidine** types.

1. **Tropane**-type alkaloids
   - Benzoylecgonine
   - Cocaine (0.3 – 1.5%; avg- 0.8%)
   - Ecgonidine
   - Ecgonine
   - Hydroxytropacocaine
   - Tropacocaine
   - Methylecgonine cinnamate
   - truxiline
2. **Pyrrolidine**-type alkaloids
   - Cuscohygrine
   - Dihydrocuscohygrine
   - Hygrine
   - Nicotine

Cocaine



•Formula- $C_{17}H_{21}NO_4$
•Molecular weight - 303.353 g/mol
•Cocaine is a weak Base with pK of 8.6
•Usually titrated with HCl to form cocaine hydrochloride

- Causes strong psychological dependency but no or little physical dependency

- Fatal dose:
  - ✓ Orally – 1.5 gm
  - ✓ Parenterally – 1 gm
- Fatal period:
  - ✓ Upto 30 minutes

•Cocaine has pockets with both hydrophilic and lipophilic efficiency.

•This results in easy crossing of BBB (Blood Brain Barrier), thus resulting in rapid CNS stimulation.

Cocaine

- It is a stimulant, an appetite suppressant, and a **nonspecific voltage gated sodium channel blocker**, which in turn causes it to produce anaesthesia at low doses.

- Biologically, cocaine acts as a serotonin–norepinephrine–dopamine reuptake inhibitor, also known as a triple reuptake inhibitor **(TRI).**

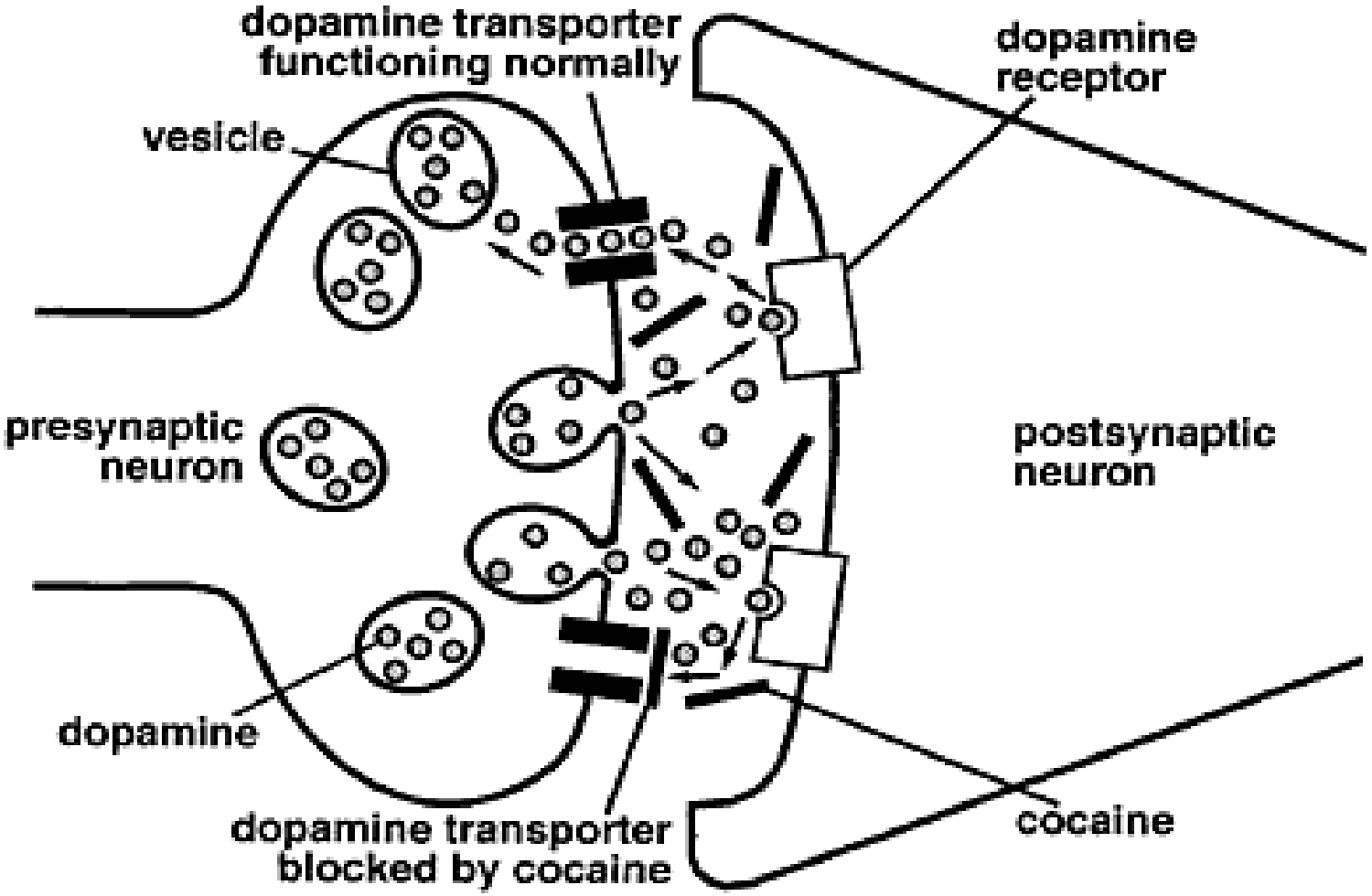
- It is addictive due to its effect on the mesolimbic reward pathway.

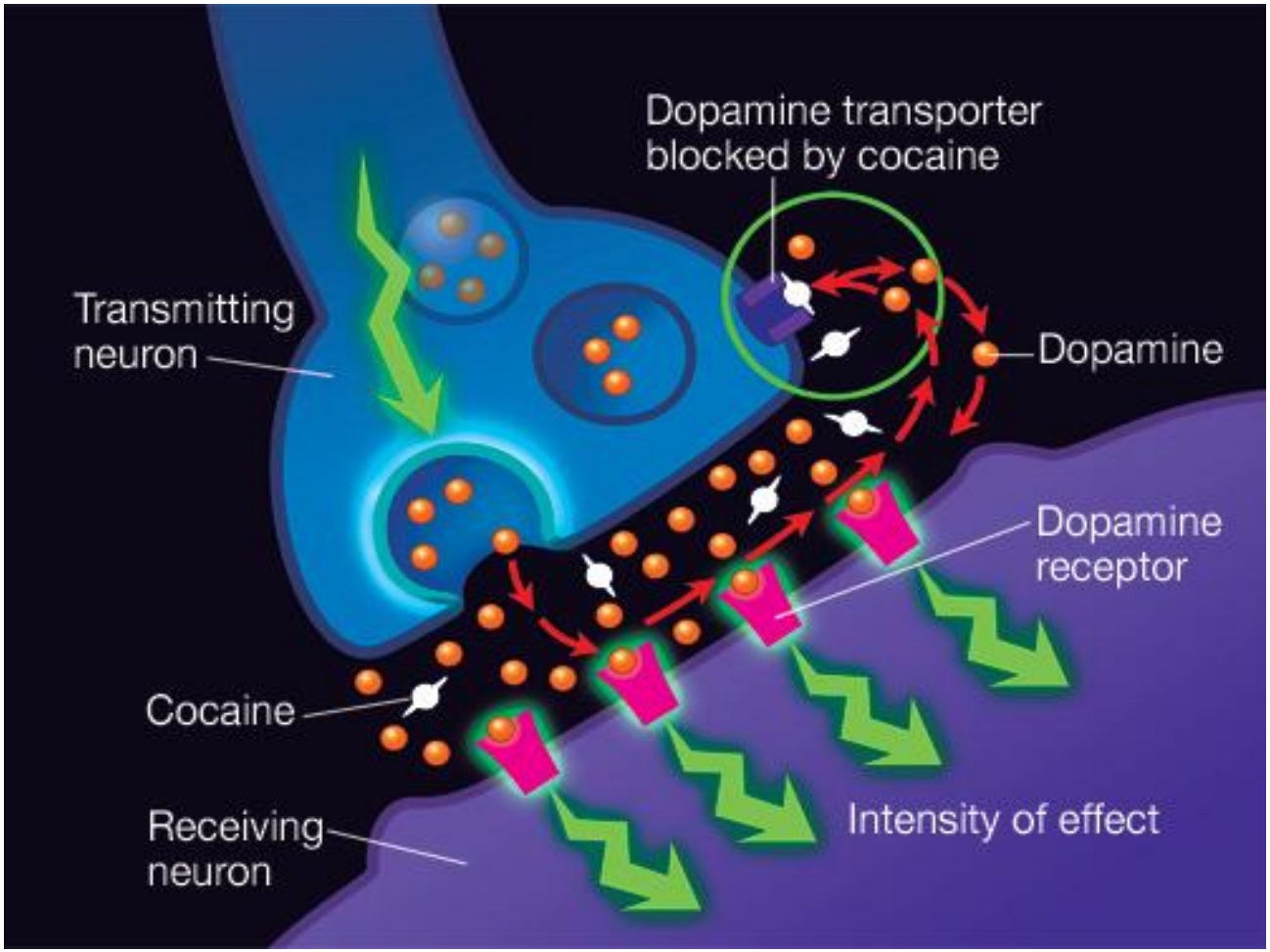
- At high doses, it is markedly more dangerous than other CNS stimulants, > **amphetamine drug class**, due to its effect on sodium channels.

Mechanism of Action

- Cocaine causes more **intense actions of neurotransmitters** unlike amphetamines which causes the increased amount of their secretions.

- It is caused by **preventing the re-uptake of neurotransmitter into presynaptic vesicles**, resulting in prolonged stay of neurotransmitter within synapse which again causes the prolonged & continuous propagation of nerve impulse which otherwise would have ceased by now.


dopamine transporter
functioning normally
dopamine
receptor
vesicle
presynaptic
neuron
postsynaptic
neuron
dopamine
dopamine transporter
blocked by cocaine
cocaine



•Cocaine binds to the dopamine re-uptake receptors and thus causes the blockage of dopamine re-entry in presynaptic membrane.

•Dopamine stays within synapses and exerts its effect for prolonged period

Signs & Symptoms

- Cocaine increases-
  - Alertness,
  - Feelings of well-being & euphoria,
  - Energy and temperature
  - Motor activity,
  - Feelings of competence,
  - Sexuality.
- Other side effects include anxiety, paranoia, restlessness, and tooth grinding (bruxism)
- Lack of sleep, itching, tachycardia, hallucinations, and paranoid delusions.
- Possible lethal side effects include rapid heartbeat (tachycardia), tremors, convulsions, markedly increased core temperature, heart attack, stroke and heart failure.
- Cocaine is highly **pyrogenic**. Heat generation is increased due to higher activity/stimulation of muscles. At the same time heat loss is inhibited by vasoconstriction. These things leads to hyperthermia.
- Studies have shown that cocaine usage during pregnancy triggers premature labor and may lead to abruption of placentae.

Signs & Symptoms

- Acute poisoning
  - Excitement, restlessness, dilated pupils, palpitation
  - Mania, convulsions, tremors
  - Collapse, respiratory and cardiac failure
  - Death

- Cocainism/ cocainomania/chronic cases
  - Rundown of general health, digestive disturbances
  - Blackish pigmentation of tongue,
  - Ulceration of nasal septum (sniffing)
  - Dementia (degenerative changes in CNS)
  - **Magnan's syndrome**; tactile hallucinations with crawling or creeping sensations as if due to "insects/bugs"

Color Tests

- Scott Test:
  - Reagent – 2% cobalt thiocyanate solution, HCl, chloroform
  - Observation – blue colour

- Gold Chloride Test:
  - Reagent – 5% $AuCl_2$ solution in water
  - Observation – precipitate is formed with **delicate rosette** or **long rod** shaped crystal appearing under microscope
  - Precipitate is insoluble in HCl

Color Tests

- **Color tests** are **simple, rapid, qualitative chemical tests** used in forensic chemistry to provide a **preliminary indication** of the presence of specific chemical classes of compounds (such as drugs, poisons, or explosives) by observing **color changes** that occur when the sample reacts with particular reagents.

- **Purpose**
  - To obtain **presumptive (screening) results** before confirmatory instrumental analysis.
  - To quickly **narrow down the class of compound** or substance category.
  - Commonly used in **drug analysis**, **toxicology**, and **trace evidence examination**.

Basis of Color Tests

- Color tests are based on **specific chemical reactions** between the analyte and reagents that lead to the formation of **colored complexes or products**.

- **Fundamental Principle**
  - The reagent reacts with certain **functional groups** in the test substance (e.g., phenols, amines, alkaloids, carbonyls).
  - These reactions produce a **visible color change** characteristic of that chemical class.

General Procedure for a Color Test

- **Step 1 — Sample Preparation**
  - Use a **small amount (1–2 mg or a few drops)** of the suspected material.
  - Dissolve in suitable solvent if required (water, ethanol, or dilute acid).

- **Step 2 — Addition of Reagent**
  - Add **1–2 drops of reagent** specific to the test.
  - Add **next reagent** (if multiple reagents) and mix well.

- **Step 3 — Observation**
  - Observe and record **color change**, **precipitate formation**, or **fluorescence**.
  - Note **color intensity**, **time of color development**, and **stability** of color.
  - Compare with **known reference standards** or **color charts**.
- **Step 4 — Interpretation**
  - Match color reaction with known responses of target substances.
  - If positive, forward sample for **confirmatory instrumental analysis**.

Forensic Significance of Color Tests

Aspect	Importance
Speed	Rapid screening tool for on-scene or laboratory use.
Simplicity	Requires minimal equipment and training.
Cost-effectiveness	Inexpensive preliminary analysis method.
Field Application	Useful in roadside drug testing, customs screening, or scene investigation.
Laboratory Utility	Guides confirmatory testing pathway and evidence prioritization.
Legal Relevance	Admissible only as presumptive evidence, not for conviction.

TLC

- **Solvent system** –
  1. Chloroform : Dioxane : Ethyl Acetate : 29%Ammonia – 25:60:10:15
  2. Methanol : 29%Ammonia – 100:1.5

- **Spraying reagent –**
  - Dragendorff's reagent- orange/brown coloured spots
  - Acidified iodoplatinate reagent- violet/bluish coloured spots

- **HRf values** of Cocaine are **81** & **59** for system1 & system2 respectively

Opioids

Papaver somniferum

- **Common names-** poppy, white poppy, opium poppy, carnation poppy
- **Opium** is air-dried, milky exude obtained by incising the unripe capsules(pods) of poppy plant.
- **Morphine** is the most abundant alkaloid among the more than 45 alkaloids present in opium.
- Traces of morphine are present in ripe & dry pods but other alkaloids aren't present in ripe pods.





Constituents of Opium

- Morphine- 10 - 20%
- Narcotine- 0.75 - 10%
- Papaverine- 0.5 - 1%
- Thebaine- 0.2 - 1%
- Codeine- 0.2 - 0.8%
- Narceine- 0.1 - 0.5%

OPIOIDS

- **Natural opioids:**
  - Opium (morphine, codeine, thebaine)
- **Semisynthetic opioids:**
  - Heroin & hydromorphone (morphine) and oxycodone & etorphine (thebaine)
- **Synthetic opioids:**
  - Fantanyl, methadone, pentazocine, meperidine etc.
- **Endogenous opioids:**
  - Endorphines

Mechanism of Action

Opioid Receptor	Function	Effects When Stimulated
μ (Mu) receptors	Main analgesic and addictive effects	Analgesia, euphoria, respiratory depression, miosis, physical dependence
κ (Kappa) receptors	Spinal analgesia	Sedation, dysphoria, psychomimetic effects
δ (Delta) receptors	Emotional response	Modulates analgesia and mood

Clinical features in opium consumption

- Affects the CNS and produces the altered state of mind.
- Relieves the pain and induces the sleep.
- Initial excitement and state of well being, followed by drowsiness and sleep.
- The effects include euphoria, drowsiness, nausea, itching, constipation and respiratory depression.
- Slow and shallow breathing, and coma.
- Causes the physiological & psychological dependency.

➢ **Withdrawal symptoms:** watery eyes, running nose, loss of appetite, yawning, irritability, tremors, chills and sweating

Signs and Symptoms Opioid Abuse

Long-term use leads to profound **neuroadaptation** and physical/psychological dependence.

System	Symptoms
CNS	Sedation, stupor, coma
	Tolerance, dependence, cognitive impairment
GI	Severe constipation, abdominal pain
Cardiac	Bradycardia, hypotension
Hormonal	↓ testosterone, menstrual irregularities
Skin	Cold skin, Track marks (IV users), infections
Respiratory	Respiratory depression, Hypoventilation, cyanosis
Immune	Reduced immunity

Behavioral and Psychological Signs

- Drug-seeking behavior
- Irritability, anxiety
- Loss of interest in daily activities
- Social withdrawal
- Mood swings
- Financial or legal problems

Morphine

- Chief constituent of opium
- Was isolated from opium in 1806 by Serturne.
- 10-20% of all opium alkaloids.
- Name is derived from Greek god Morpheus; the god of dreams.
- A phenanethrene alkaloid ($C_{17}H_{19}O_3N$).
- White powder or white shinning crystals form with bitter taste.
- **Fatal dose:**
  - 200-250mg orally
  - 50-100mg injection

Heroin

- **Slangs:** Diacetyl morphine, horse, smack, junk, dope, brown sugar.

- Sniffed, smoked and injected.
- White sugar is pure heroin, while brown sugar (illicit heroin) contains diluents like cocoa, starch, milk powder etc.
- Four times more powerful in action than morphine.
- The euphoria produced is higher than any other narcotics.
- The effect of heroin is felt within one minute after injection. Intense euphoria lasts for several minutes followed by sedation.
- There is feeling of warmth and intense pleasure.
- Causes physical and physiological dependency.

- **Withdrawal symptoms** include painful abdominal cramps, nausea, vomiting, restlessness, congestion and altered heart rate.

Colour tests for Morphine

- **Frohde's Test:**
  - To the dried residue of extract is added one drop of Frohde's reagent(1% ammonium molybdate solution in concentrated H_2SO_4).
  - A violet colour changing to green and finally blue is observed.
- **Marquis Test:**
  - To the dried residue of extract is added one drop of Marquis reagent (one drop of formalin in one ml of concentrated H_2SO_4).
  - A purple red colour is produced which changes to violet and blue.

- **Zernic Nitric Acid Test**:
  - To the dried residue of extract is added one drop of concentrated HNO_3.
  - Presence of yellow colour is indication of opium alkaloids.
- **Urotropine Test:**
  - To the dried residue of extract is added a few drops of aqueous solution of urotropine and warmed slightly.
  - A purple colour changing to blue and then green is observed.

Alkaloid	Colour observed in Marquis test	Colour observed In Mecke test	Colour observed In Frohde's test	Colour observed In nitric acid test
Opium	Purple-violet	Deep green	Purple	Yellow
Morphine	Purple-violet	Deep green	Purple	Orange
Codeine	Purple	Bluish green	Blue	Orange
Thebaine	Orange	Orange	Blue to green	Yellow
Heroin	Purple –violet	Deep green	purple	Yellow

Microcrystal Tests For Morphine

- These tests are performed with different reagents on the residue of extract.
- The crystals formed show definite geometry, characteristics to a particular plant poison.
- These tests may be used for screening f the samples.

Reagent	Type of crystal
5% mercuric chloride	Tufts
Potassium mercuric iodide	Needles
5% potassium iodide	Orange plates

TLC of Opium Alkaloids

- **Solvent system:**
  - Carbon tetrachloride : n-butanol : methanol : 6N ammonia (40:30:3:2)
  - Xylene : butanone : methanol : diethyleamine (40:40:6:2)
- HRf values of opium alkaloids-

Active component	Sys 1	Sys 2
Narcotine	90	74
Papaverine	85	70
Thebaine	80	56
Codeine	60	37
Morphine	40	15
Narceine	25	02

Gas Chromatographic Analysis of Opioids

- The purified alkaloids from TLC plate is dissolved in 0.5 ml alcohol and aliquot of 5 microlitre is injected in GLC column.
- Method-
  - Column: length 3 ft, diameter: 1/8”
  - Material: spiral glass
  - Stationary phase: 1% high efficiency- 8 B
  - Carrier Gas: Nitrogen with flow rate of 60ml/min
  - Temp.: 220C
  - Detector: FID

UV and IR Spectrometry of Opioids

- Principal peaks of morphine in IR spectra
  - 805, 1243, 1118, 945, 1086, 833

- λmax (in nm) in diverse media (UV spectrography)
  - Aqueous acid : 285
  - Aqueous alkaline : 298

Hallucinogens

- Hallucinogens are **psychoactive substances** that alter perception, mood, and thought processes, often leading to **visual and auditory hallucinations**.

- **Common Hallucinogens**
  - LSD (Lysergic Acid Diethylamide)
  - Psilocybin (Magic Mushrooms)
  - PCP (Phencyclidine)
  - MDMA (Ecstasy)
  - Mescaline (Peyote)
  - Ketamine
  - DMT (Dimethyltryptamine)

Hallucinogens

Psychological Effects	Physical Effects	Severe Effects
• Hallucinations (visual, auditory, or tactile) • Distorted perception of time and space • Paranoia and delusions • Extreme mood swings (euphoria to anxiety) • Depersonalization (feeling detached from self) • Impaired judgment and reasoning	• Dilated pupils • Increased heart rate (tachycardia) • High blood pressure (hypertension) • Increased body temperature • Sweating or chills • Nausea or vomiting • Poor coordination and dizziness	•Panic attacks and extreme paranoia •Seizures •Coma (in case of severe overdose, e.g., PCP) •Persistent psychosis (rare but possible with prolonged use) •Hallucinogen Persisting Perception Disorder (HPPD – **flashbacks** of hallucinations after the drug has worn off)

Depressants

- **Depressants** are drugs that **reduce the activity of the central nervous system (CNS)** by enhancing inhibitory neurotransmission, resulting in **sedation, relaxation, decreased anxiety, and drowsiness**.
- Also known as sedative-hypnotics or tranquilizers.
- At higher doses, they can cause respiratory depression, coma, and death.
- Commonly misused for recreational, self-medication, and criminal purposes (e.g., drug-facilitated sexual assault).
- **Common Depressants**
  - Benzodiazepines (e.g., Diazepam, Alprazolam)
  - Barbiturates (e.g., Phenobarbital)
  - GHB (Gamma-Hydroxybutyrate)

Classification of Depressants

Category	Examples	Clinical Use
Barbiturates	Phenobarbital, Pentobarbital, Secobarbital	Anesthesia, epilepsy, insomnia
Benzodiazepines	Diazepam, Lorazepam, Alprazolam, Rohypnol	Anxiety, insomnia, muscle relaxation, seizure control
Other CNS Depressants	GHB (Gamma-Hydroxybutyrate), Chloral hydrate, Meprobamate	Sedative, “club drug” use

Mechanism of Action

- All these depressants **enhance the activity of the inhibitory neurotransmitter GABA (Gamma-Aminobutyric Acid)** at the **GABA_A receptor complex** in the brain.

- GABA binds to GABA_A receptors → opens **chloride ion channels** → hyperpolarization of neurons → **decreased neuronal excitability**.

- Result: **Sedation, anxiolysis, hypnosis, muscle relaxation, anticonvulsant effects.**

Depressants

Psychological Effects	Physical Effects	Severe Effects
•Sedation and drowsiness •Slowed cognitive function •Poor concentration and confusion •Mood changes (euphoria or depression) •Memory impairment (especially with benzodiazepines and alcohol)	• Slowed breathing (respiratory depression) • Lowered heart rate (bradycardia) • Low blood pressure (hypotension) • Poor motor coordination and slurred speech • Reduced reflexes and response time	•Respiratory depression (can be fatal) •Unconsciousness or coma •Extreme confusion or agitation •Death (especially with opioids like fentanyl or barbiturate overdose)

Stimulants vs Depressants

Aspect	Stimulants	Depressants
Definition	Drugs that increase brain activity, alertness, and energy levels.	Drugs that slow down brain activity and reduce arousal.
Examples	Cocaine, amphetamines, caffeine, nicotine.	Alcohol, benzodiazepines, barbiturates.
Mechanism	Increase neurotransmitter release (e.g., dopamine, norepinephrine).	Enhance the action of inhibitory neurotransmitters (e.g., GABA).
Effects	Increased heart rate, blood pressure, alertness, and energy.	Relaxation, sedation, slowed reflexes, and decreased heart rate.
Medical Uses	Treatment of ADHD, narcolepsy, and obesity.	Treatment of anxiety, insomnia, and seizures.
Side Effects	Anxiety, paranoia, insomnia, cardiovascular issues.	Drowsiness, impaired motor skills, respiratory depression.

Stimulants vs Hallucinogens

Aspect	Stimulants	Hallucinogens
Definition	Drugs that increase brain activity and energy.	Drugs that alter perception, mood, and cognition.
Examples	Cocaine, amphetamines, caffeine.	LSD, psilocybin (magic mushrooms), PCP, mescaline.
Mechanism	Act on dopamine, norepinephrine, and serotonin pathways to increase stimulation.	Interfere with serotonin receptors to alter sensory perception.
Effects	Increased alertness, energy, euphoria, and physical stamina.	Distorted reality, hallucinations, and altered sense of time and space.
Medical Uses	ADHD, narcolepsy (e.g., amphetamines).	Limited use (e.g., psilocybin for mental health in research).
Side Effects	Addiction, insomnia, cardiovascular issues.	Anxiety, psychosis, "bad trips," and flashbacks.

Stimulants vs Depressants vs Narcotics

Aspect	Stimulants	Depressants	Narcotics
Definition	Increase brain activity and alertness.	Slow down brain activity and induce relaxation.	Relieve pain and produce euphoria.
Examples	Cocaine, amphetamines, caffeine.	Alcohol, barbiturates, benzodiazepines.	Morphine, heroin, codeine, oxycodone.
Mechanism	Increase release of dopamine and norepinephrine.	Enhance GABA action (inhibitory).	Act on opioid receptors in the brain to block pain signals.
Effects	Alertness, energy, euphoria, increased heart rate.	Sedation, relaxation, reduced heart rate, and slowed reflexes.	Pain relief, drowsiness, euphoria, respiratory depression.
Medical Uses	ADHD, narcolepsy.	Anxiety, insomnia, seizures.	Pain management (e.g., post-surgical pain).
Side Effects	Insomnia, paranoia, cardiovascular risks.	Respiratory depression, memory problems, dependency.	Addiction, respiratory depression, overdose.

Short-Term vs Long-Term Impacts

Drug Class	Short-Term Signs & Symptoms	Long-Term Signs & Symptoms
Stimulants	Increased heart rate, alertness, euphoria, restlessness, insomnia.	Anxiety, paranoia, psychosis, weight loss, heart problems.
Depressants	Relaxation, sedation, drowsiness, slowed reflexes.	Dependency, memory problems, tolerance, respiratory issues.
Narcotics	Pain relief, euphoria, drowsiness, slowed breathing.	Addiction, liver damage, tolerance, respiratory depression.
Hallucinogens	Altered sensory perception, hallucinations, anxiety.	Flashbacks, persistent psychosis, mood disorders.

NDPS Act, 1985

1. Introduction to NDPS Act, 1985

The **Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985**, is a comprehensive law enacted by the Government of India to regulate and control the production, manufacture, possession, transport, sale, purchase, use, and consumption of **narcotic drugs** and **psychotropic substances**. The Act also aims to prevent the illicit trafficking of these substances while ensuring that legitimate medicinal and scientific needs are met.

Enactment and Scope:

- Enforced on **14th November 1985**.
- It prohibits the cultivation, production, trade, and use of narcotic drugs and psychotropic substances, except for medical or scientific purposes.
- Extends to the entire territory of India and also applies to Indian citizens and ships or aircraft registered in India outside its territory.

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2. Objectives of the NDPS Act, 1985

1. **Control and Regulation**:
    - Prevent and control the misuse of narcotic drugs and psychotropic substances.
    - Regulate the production, manufacture, possession, transport, import, export, sale, and consumption of such substances.
2. **Prevent Illicit Trafficking**:
    - Dismantle organized drug cartels and trafficking networks.
    - Control the flow of illicit drugs both domestically and internationally.
3. **Ensure Medical Use**:
    - Facilitate the legitimate use of narcotics and psychotropics for medical and scientific purposes.
4. **Deterrence and Punishment**:
    - Impose stringent punishments for drug-related offenses, including cultivation, trafficking, and consumption.
5. **International Obligations**:
    - Align with international conventions on drug control, such as the Single Convention on Narcotic Drugs, 1961.

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3. Major Sections in NDPS Act

Sections 16–22: Offenses and Penalties

Section	Description
Section 16	Punishment for contravening orders related to **poppy straw**. Punishment: Up to **10 years imprisonment** and fine.
Section 17	Penalty for production, manufacture, possession, or trafficking of **prepared opium**. Punishment: Varies with the quantity involved.
Section 18	Penalty for illegal cultivation of **opium poppy, cannabis, or coca plants**. Punishment: Up to **10 years imprisonment** and fine.
Section 19	Punishment for those who **embezzle opium** or opium derivatives licensed under the Act. Penalty: Up to **10 years imprisonment**.
Section 20	Punishment for cultivation, production, possession, trafficking, or consumption of **cannabis**: - Small Quantity: Up to **1 year imprisonment or fine of Rs. 10,000 or both**. - Commercial Quantity: Rigorous imprisonment of **10–20 years** and fine between **Rs. 1–2 lakh**.
Section 21	Punishment for manufacturing, possession, sale, or transport of **manufactured drugs or preparations**.
Section 22	Punishment for contravention related to **psychotropic substances**: - Small Quantity: Up to **1 year imprisonment or fine of Rs. 10,000 or both**. - Commercial Quantity: Rigorous imprisonment of **10–20 years** and fine between **Rs. 1–2 lakh**.

Section 27: Punishment for Personal Use (Consumption)

- **Punishment**:
  - For **cocaine, morphine, heroin**: Up to **1 year imprisonment** or fine of Rs. 20,000 or both.
  - For other drugs: Up to **6 months imprisonment** or fine of Rs. 10,000 or both.
- **Rehabilitation**: The court may recommend **de-addiction treatment** instead of punishment if the offender is an addict.

Section 31: Enhanced Punishment for Repeat Offenders

- If the offender commits any offense under the Act after a prior conviction, **enhanced punishment** will be imposed:
  - **Double the maximum term of imprisonment** prescribed for the offense.
  - Fine: **Double the maximum amount** prescribed.
  - For certain offenses, the death penalty can be imposed in cases of repeat offenders involved in large-scale trafficking of drugs.

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4. Medico-Legal Aspects of the NDPS Act

- **Role in Forensics**:
    - Helps law enforcement and forensic experts investigate drug-related crimes.
    - Forensic analysis determines the type and quantity of drug involved, which impacts penalties.
- **Burden of Proof**:
    - Prosecution must prove the intentional possession of drugs by the accused.
- **Addicts and Rehabilitation**:
    - Special provisions allow courts to direct addicts to **de-addiction centers** rather than sentencing them to prison.
- **Confiscation of Property**:
    - Assets acquired through illicit drug trafficking can be confiscated under the Act.

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5. Challenges

1. **Illicit Cultivation**: Hidden or disguised drug crops in remote areas.
2. **International Trafficking**: India's geographic location makes it vulnerable to trafficking.
3. **Legal Loopholes**: Challenges in proving "possession with intent" in courts.
4. **Emerging Threats**: Synthetic drugs and designer drugs not initially covered under the Act.

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6. Relevance in Forensic Chemistry

- **Forensic Role in NDPS Cases**:
    - Identify narcotic drugs and psychotropic substances through chemical and instrumental methods.
    - Quantify the **small** or **commercial quantity** for legal proceedings.
    - Use of advanced techniques like **GC-MS, HPLC, and FTIR** for drug profiling.

Conclusion

The NDPS Act, 1985, is a cornerstone of India's drug control framework. Its stringent provisions aim to curb the menace of drug abuse while balancing the need for legitimate medical and scientific use. Understanding the **legal provisions**, **medico-legal aspects**, and **forensic applications** is essential for students of forensic chemistry.